BOAMP M0 Procurement Recurrence Linking

Detailed Technical Reference for Data Audit, Preprocessing, Recurrence-Linking Algorithm, and Survival Analysis

Gigalis Predictive-Modeling Internship — Current Independent Study

Repository state of July 2026 (outputs last regenerated 2026-07-15 via the notebook front-ends; key datasets byte-identical to the audited 2026-07-13 script run)

Abstract

This report is the detailed technical reference for the current BOAMP-only procurement recurrence-linking pipeline covering digital/ICT public contracts in Pays de la Loire, 2015–2026. It documents the concrete implementation choices, data audit, preprocessing rules, the deterministic M0 recurrence-linking algorithm, intermediate diagnostics, and survival-modeling outputs, in a form that can be checked, reproduced, and handed off. BOAMP notices carry no field identifying whether a call for tenders (*appel d'offre*) is a recurrence of an earlier contract for the same buyer; the relationship must be inferred from observable signals: buyer identity, free-text object similarity, CPV classification, and timing relative to the source contract's estimated end date.

Starting from 84,623 cleaned BOAMP notices published for Pays de la Loire buyers between 2015-03-02 and 2026-07-13, the pipeline identifies 3,159 digital/ICT `APPEL_OFFRE` source notices. A deterministic four-component scoring rule (text, CPV, timing, buyer-key reliability) is applied to 6,137 blocked candidate pairs; at the reference ("balanced") threshold this yields 618 linked proxy-recurrence events, an event rate of 19.6%. The official survival handoff, `boamp_survival_m0_balanced.csv`, contains one row per eligible source notice with a binary `event` indicator and an observed `time_to_event_or_censor_months` survival time.

Kaplan–Meier survival never crosses 50% under the reference definition (median survival is undefined; restricted mean survival time at 60 months is 53.5 months), reflecting 80.4% censoring. A buyer-clustered Cox model finds a statistically significant negative association between the duration-imputation flag and the recurrence hazard (HR = 0.433, $p < 0.001$), while the effect of declared duration itself is small and not significant in the reference design (HR = 0.923, $p = 0.38$). A dedicated duration-leakage audit shows this duration effect is *not* robust: it reverses sign (HR = 1.344) under an alternative linkage design that does not center candidate search on declared duration, indicating that duration-related survival findings are conditional on the current temporal-blocking design rather than independent substantive effects. Blocking coverage is a first-order limitation: only 39.1% of eligible sources have at least one scored candidate. The reference event set should therefore be read as a *proxy recurrence signal* constructed under a fully documented, auditable rule — not as a verified renewal probability.

How to read this report

This document is the single, comprehensive technical reference for the current repository state. It is organized the way a handoff document should be: Section 1 gives the business framing, the objectives, and the research questions; Section 2 documents the computational workflow and run flow as they actually exist in the repository, including where implementation is duplicated between scripts and notebooks; Sections 3–5 document the raw data and its quality; Sections 6–8 document preprocessing and the M0 linking algorithm mechanically, with a full worked example in Section 10; Section 12 consolidates the credibility diagnostics (blocking coverage, score margins, an unsupervised Fellegi–Sunter precision/recall estimate, corruption-recovery robustness); Sections 13–15 document the survival analysis, its methods, results, and robustness under alternative event definitions; Section 17 classifies every method by its implementation status (verified / pending validation / partial / deferred / proposed); Section 18 states plainly what is and is not supported by the current evidence. All figures and tables reflect the current repository run; every number quoted here is reproduced in a machine-readable CSV under `reports/tables/` or `data/processed/`, referenced inline.

Relationship to the other documents in `reports/`. This report is the master reference and subsumes the others. `boamp_m0_preprocessing_report.md` is the standalone data-retrieval and preprocessing report (Sections 3–6 here, in more operational detail); `linkage_quality_evaluation.tex/.pdf` is the focused credibility-audit companion (Sections 9–15 here); `credibility_audit_report.md` is its short machine-checkable executive summary; `run_logs/run_log.md` is the append-only execution history. Where documents overlap, they were checked for numerical consistency against the same generated tables; no document reports a value the others contradict.

Epistemic labeling convention. Throughout, claims are separated into four kinds: (i) *literature-standard* methods, cited to their sources in the bibliography; (ii) *project-specific methodological choices* (e.g. score weights, threshold percentiles), which are documented decisions, not fitted or externally justified quantities; (iii) *data- or compute-constrained* choices (e.g. TF-IDF instead of transformer embeddings, the Pays-de-la-Loire re-scoping), where the constraint is stated; and (iv) *proposed improvements*, which are not implemented and are listed only in Sections 17 and 19.

Contents

1 Project Scope

1.1 Business Context

Gigalis manages a portfolio of public digital-infrastructure contracts in the Pays de la Loire region. A recurring operational question is: *when is a given procurement contract likely to be re-tendered?* Anticipating this lets account managers prepare competitive offers ahead of a renewal campaign instead of discovering a re-tender after the fact.

1.2 Why BOAMP?

The *Bulletin Officiel des Annonces de Marchés Publics* (BOAMP) is the official French public-procurement gazette. Contracts above the relevant thresholds must be published there, across the notice lifecycle: pre-information, call for tenders (`APPEL_OFFRE`), award (`ATTRIBUTION`), and correction/modification notices.

1.3 Core Problem

BOAMP does not provide an explicit field linking an original call for tenders to its later recurrence. Each notice is a standalone publication event. The recurrence relationship must be **inferred** from a combination of signals: the same buyer re-issuing a broadly similar procurement need in the time window around when the original contract was expected to end.

1.4 Scope of the Current Study

This is a fresh, independent pipeline; it does not reuse or compare against any prior BOAMP study, report, or event count. It targets a specific, documented scope bundle:

- **Period**: 2015-01-01 to 2026-07-13 (latest available). The BOAMP Opendatasoft corpus itself begins 2015-03-02, so January–February 2015 contribute no notices by construction, not by retrieval failure.
- **Geography**: Pays de la Loire only — buyer department code in $\{44, 49, 53, 72, 85\}$ (Loire-Atlantique, Maine-et-Loire, Mayenne, Sarthe, Vendée), applied **server-side** at retrieval time via the API's `where` clause.

- **Theme**: digital/ICT contracts — CPV divisions $\{32, 35, 48, 72\}$ (telecommunications equipment, security, software, IT services) and their subcategories, applied as a **hard filter** when constructing the M0 source population (Section 6).

No external SIREN/SIRET enrichment is performed anywhere in this pipeline: no INSEE SIRENE API call, no data.gouv.fr entreprise search, no external company database. Every identifier used is a BOAMP-provided value that is only format/checksum-validated, never looked up or inferred externally (Section 6.1).

1.5 Objectives and Research Questions

The study's operational objective is a documented, reproducible, BOAMP-only pipeline that turns raw procurement notices into a contract-level survival dataset suitable for time-to-recurrence modeling. The research questions actually addressed by the current implementation are:

RQ1. *Can recurrence of a procurement need be operationalized from BOAMP alone?* — i.e. can a transparent, deterministic rule link an initial call for tenders to a plausible later re-tender by the same buyer, without external enrichment or supervised learning? (Sections 6–8.)

RQ2. *How reliable are the BOAMP fields the rule depends on* — buyer identifiers, CPV codes, and especially the declared contract duration? This includes the internship guide's Week-1 "scientific question" on duration-field reliability. (Sections 3–5.)

RQ3. *What does the time-to-recurrence distribution of the constructed proxy event look like*, and which observable covariates associate with the recurrence hazard? (Sections 13–14.)

RQ4. *How credible and how stable are these answers* under internal diagnostics (model-based precision/recall, corruption stress tests) and under perturbations of the event definition itself (thresholds, temporal windows, duration-independent designs)? (Sections 12, 15.)

The short answers, defended in the body of the report, are: RQ1 — yes, mechanically, but only as a *proxy* signal whose external accuracy is unmeasured; RQ2 — buyer identifiers and durations are weak (72.8% name-fallback keys; 88.0% imputed durations), CPV is moderate and decaying in recent years; RQ3 — the proxy recurrence is a slow event (median survival undefined; RMST_{60} = 53.5 months) whose strongest apparent covariate effects are entangled with the linkage design; RQ4 — internally coherent but materially sensitive to the temporal design, with a demonstrated duration-circularity artifact that is the report's highest-priority caveat.

2 Computational Workflow and Run Flow

2.1 Observed Pipeline Structure

The repository implements a linear data pipeline followed by a fan-out of evaluation analyses. The persisted datasets and their producer scripts are inventoried in Table 6 (Section 4); the execution order, reconstructed from script input/output declarations and `reports/tables/audit_data_lineage.csv`, is:

1. **Retrieval** (`scripts/download_boamp.py`): DILA Opendatasoft API → `data/raw/boamp/pdl/boamp_YYYYMM.json` (139 monthly files + a download manifest; raw files are never overwritten).
2. **Flattening** (`scripts/parse_boamp.py`): raw JSON → `data/interim/boamp_raw_flattened.csv` (84,623 notices × 42 columns), via the adaptive schema extractor in `src/utils/boamp_schema.py`.

3. **Preprocessing** (notebook 05 / script preprocess_boamp_m0.py): cleaned notice table (boamp_clean_m0_no_enrichment.csv, 93 columns) and the digital-scope source population (boamp_m0_sources.csv, 3,159 rows).

4. **Candidate generation and scoring** (notebook 07 / script build_m0_candidate_pairs.py): blocked, scored pairs (boamp_m0_candidate_pairs.csv, 6,137 rows).

5. **Linkage and survival handoff** (notebook 08 / script run_m0_linkage.py): broad/balanced/strict link files and the reference survival dataset (boamp_survival_m0_balanced.csv), plus the m0_*.csv quality tables.

6. **Evaluation fan-out** (consumes stages 3–5, never mutates them): repository audit (audit_current_project.py / notebook 13), Fellegi–Sunter mixture (notebook 11 / script eval_m3_*), corruption recovery (notebook 12 / eval_m4_*), threshold/ablation/variant-KM sensitivity (eval_m6a/b/c_*, script-only), temporal-window sensitivity (notebook 14 / eval_m6d_*), duration leakage (notebook 15 / eval_duration_leakage.py), and the survival analysis (notebook 16 / run_survival_analysis.py). Sensitivity runs write their own link/survival datasets (boamp_m0_links_window_*m.csv, boamp_survival_m0_*.csv) — these are sensitivity artifacts, not replacements for the balanced reference handoff.

7. **Reporting utilities**: make_figures.py, make_evaluation_figures.py, make_report_figures.py (figures 01–18), and the report-assembly helpers _build_schema_summary.py, _build_source_trace.py, _build_final_audit.py.

2.2 Dual Script/Notebook Implementation — Observed State

The pipeline exists in two forms, and the relationship between them is not uniform. This is reported as observed, not idealized:

- **Stages 3–5 are implemented twice.** Notebooks 05/07/08 contain the preprocessing, candidate generation, and linkage logic *inline* (they do not import the scripts), while scripts preprocess_boamp_m0.py, build_m0_candidate_pairs.py, and run_m0_linkage.py implement the same logic as a headless batch backend. Both write the same official output files. The duplication is a maintenance risk (a rule changed in one place could silently diverge from the other), but as of the current run the two implementations are *verified equivalent in output*: the 2026-07-15 notebook regeneration reproduced the audited 2026-07-13 script outputs byte-for-byte (SHA-256 of boamp_m0_sources.csv, boamp_m0_candidate_pairs.csv, and boamp_survival_m0_balanced.csv match reports/tables/audit_dataset_inventory.csv exactly).

- **Stage-6 notebooks 14 and 15 share code with the scripts.** They import build_m0_candidate_pairs and run_m0_linkage from scripts/, so the M0 scoring rule used in the temporal-window and duration-leakage sensitivity analyses has a single definition.

- **Notebooks 11, 12, and 16 are inline analyses** (no script import); notebook 13 is the only remaining thin viewer, with a RUN_SCRIPT toggle over audit_current_project.py.

- **Evaluation scripts m6a/m6b/m6c have no notebook counterpart**; they are script-only.

2.3 Run History

Two full pipeline runs are logged in reports/run_logs/run_log.md. **Run 1** (2026-07-13) retrieved a national, all-sector, 2024–2026 corpus (340,772 notices); it is *superseded*: its raw files were moved to data/raw/boamp_national_2024_2026_archive/ and none of its outputs remain in the active lineage. **Run 2** (2026-07-13, the current run) re-scoped retrieval to the internship guide's recommended bundle (Pays de la Loire, digital/ICT, 2015–latest) and produced every dataset and result in this report. A third event — the 2026-07-15 migration of preprocessing/linkage/analysis into the notebook front-ends and full regeneration of outputs — reproduced Run 2's results exactly but is *not* logged as a run-log entry; adding one is a suggested improvement below.

2.4 Suggested Workflow Clarifications (Not Implemented)

These are proposals, clearly separated from the observed state: (i) extract the stage-3–5 logic duplicated between notebooks `05/07/08` and their scripts into shared `src/` modules imported by both, as notebooks 14/15 already do for scoring; (ii) append a run-log entry for the notebook migration and for any future regeneration; (iii) refresh `audit_credibility_implementation_matrix.csv`, whose `status` column records the state *before* the credibility audit (several items marked "NOT IMPLEMENTED" there, e.g. the duration-leakage audit and PH diagnostics, have since been implemented — Section 17 gives the current classification); (iv) add notebook front-ends (or retire) the script-only `m6a/b/c` evaluations for consistency with the notebook-first workflow.

3 Input Data and Column Dictionary

3.1 Source and Extraction

The dataset was extracted from the **DILA Opendatasoft Explore API v2.1**, dataset `boamp` (`https://boamp-datadila.opendatasoft.com/api/explore/v2.1/catalog/datasets/boamp/`), using the `.../exports/json` endpoint (no 10,000-row offset cap) with a `select=` projection that drops the redundant `gestion` metadata block while keeping the full `donnees` payload. The geographic filter was confirmed live against the API before committing to the full download:

Table 1: Live API check motivating the geographic filter (source: `reports/boamp_m0_preprocessing_report.md`).

Query	`total_count`
January 2024, national (no filter)	8,728
January 2024, Pays de la Loire filter	531 (6.1%)
Full period 2015-01-01 to 2026-07-13, PdL filter	**84,623**

Actual retrieved: 84,623 unique notices across 139 monthly JSON files (`reports/tables/boamp_download_summary.csv`), 0 duplicates, 0 failed requests. Raw files live under `data/raw/boamp/pdl/` (140 entries on disk: the 139 monthly files plus one `download_metadata.json` manifest). JOUE-family notices use the EU eForms/UBL JSON structure from late 2023 onward; FNS/MAPA/DSP-family notices use the legacy BOAMP-XSD-derived JSON structure throughout the period. Both are handled by a single adaptive recursive extractor (`src/utils/boamp_schema.py`), not by two separate parsers.

3.2 Column Dictionary — Raw/Flattened Fields

Table 3 describes the raw and derived fields most relevant to M0, drawn from `reports/tables/boamp_source_schema_summary.csv` and `reports/tables/boamp_observed_columns.csv`. Coverage is measured over all 84,623 flattened notices.

Table 2: Notice-type composition of the 84,623 cleaned notices.

Notice type	Count	Share
`APPEL_OFFRE` (calls for tender)	58,292	68.9%
`ATTRIBUTION` (award notices)	22,560	26.7%
Other (rectificatif, modification, ...)	3,771	4.5%
Total	84,623	100%

Table 3: Selected column dictionary — raw/flattened BOAMP fields.

Field	Coverage	Description
`idweb`	100%	Unique BOAMP notice identifier (e.g. `15-28783`). Primary key, used as `notice_id`.
`objet`	100%	Free-text procurement object/title. Primary signal for TF-IDF text similarity.
`nature`	100%	Notice type: `APPEL_OFFRE`, `ATTRIBUTION`, or other (rectificatif, modification, annulation, pre-information, ...). Defines source vs. candidate pool.
`code_departement`	100%	Notice/buyer department code(s); server-filtered to $\{44, 49, 53, 72, 85\}$ at download time.
`dateparution`	100%	Notice publication date; primary temporal ordering field.
`nomacheteur`	100%	Raw buyer/contracting-authority name; source for the normalized-name buyer-key fallback.
`annonce_lie`	30.1%	Back-reference `idweb`(s) of a linked prior notice (native BOAMP cross-reference). On `ATTRIBUTION` notices, used to recover a more precise `start_date` for the corresponding `APPEL_OFFRE`.
`titulaire`	21.2%	Winning bidder name(s); populated only on `ATTRIBUTION` notices. Not used by M0 (no winner-identity enrichment).
`siret_candidates_raw` (derived)	—	14-digit SIRET-shaped values found by recursive key-pattern search inside `donnees`. Source of `buyer_siret_clean`.
`cpv_candidates_raw` (derived)	—	8-digit CPV codes found near a key containing `cpv` (legacy: `CPV.objetPrincipal.classPrincipale`; eForms: `ItemClassificationCode`). Source of `cpv_clean` and its hierarchy fields.
`duration_months_raw` (derived)	47.7%[1]	Contract duration in months, from `dureeMois`/`nbMois` (legacy) or `DurationMeasure`+`unitCode` (eForms).

Two identifier fields are derived by cleaning, not enrichment: a raw 14-digit SIRET is accepted only if it is digit-only and its Luhn checksum passes (`src/utils/identifiers.py`); a SIREN, when no valid SIRET is present, is derived as the first 9 digits of a valid SIRET (internal cleaning of a BOAMP-provided value, since SIRET = SIREN + 5-digit NIC). Across the 84,623 cleaned notices, 27.2% carry a format-valid raw SIRET, of which 99.6% also pass the checksum test.

3.3 Digital-Scope Category Distribution

Within the 3,159 M0 source notices (Section 6), the CPV division distribution is shown in Table 4. Because the digital-scope filter accepts a notice either via CPV division *or* via a digital keyword hit in its object text (Section 6), a meaningful share of sources (21.6%) carry no CPV code

[1]47.7% is the coverage among all 84,623 notices before plausibility filtering (see `duration_quality_flag` in Table 7); coverage among only the 3,159 in-scope M0 source notices is much lower (12.0% observed, Section 5).

at all, and a further 15.6% carry a CPV division outside the core $\{32, 35, 48, 72\}$ set (division 30 – office/computing machinery, 45 – construction works with an IT-related keyword hit, 79 – business services, etc.).

Table 4: CPV-division distribution among the 3,159 digital-scope M0 source notices.

CPV division	Count	Share
72 — IT services	889	28.1%
(missing CPV, kept via keyword match)	682	21.6%
48 — Software & information systems	466	14.8%
32 — Telecommunications equipment	370	11.7%
30 — Office/computing machinery	203	6.4%
35 — Security equipment	157	5.0%
45 — Construction works (keyword hit)	87	2.8%
79 — Business services (keyword hit)	59	1.9%
71 — Architectural/engineering services	56	1.8%
Other divisions (<1.6% each)	190	6.0%
Total	3,159	100%

A stricter CPV-division-only definition of digital scope (dropping the keyword fallback) gives 1,882 sources instead of 3,159 — reported in `reports/boamp_m0_preprocessing_report.md` as a sensitivity check, not the headline population, because CPV coverage among `APPEL_OFFRE` notices is well below 100% and an exact-CPV-only rule would silently drop genuine digital notices that simply lack a recoverable CPV code.

3.4 Buyer Landscape

Table 5 summarizes buyer concentration among the 3,159 M0 sources. A relatively small number of large, repeat buyers dominates the volume: the top five buyers alone account for 457 of 3,159 sources (14.5%).

Table 5: Buyer concentration among the 3,159 M0 source notices.

Metric	Value
Total unique buyer keys	790
Buyers identified by a raw BOAMP SIRET	713 sources (22.6%)
Buyers identified by normalized name only	2,446 sources (77.4%)
Buyers with ≥ 2 digital AO sources	404
Buyers with ≥ 5 digital AO sources	132
Top 5 buyers by number of digital AO sources	
Nantes Métropole (name-keyed)	182
SIRET:23440003400026	75
Conseil régional des Pays de la Loire (name-keyed)	70
CHU de Nantes (name-keyed)	69
Angers Loire Métropole (name-keyed)	61

Figure 1 shows how the balance between SIRET-keyed and name-fallback-keyed buyers shifts across the study period. Name-fallback dominance (72.8% of all 84,623 notices, 77.4% of M0 sources specifically) is the single largest structural risk for buyer-group fragmentation (Section 5, item 2): a buyer appearing under two spellings ("Nantes Métropole" vs. "NANTES METROPOLE") is treated as two distinct buyer keys unless a raw SIRET recovers the true identity, and this project performs no fuzzy-name or external-registry reconciliation.

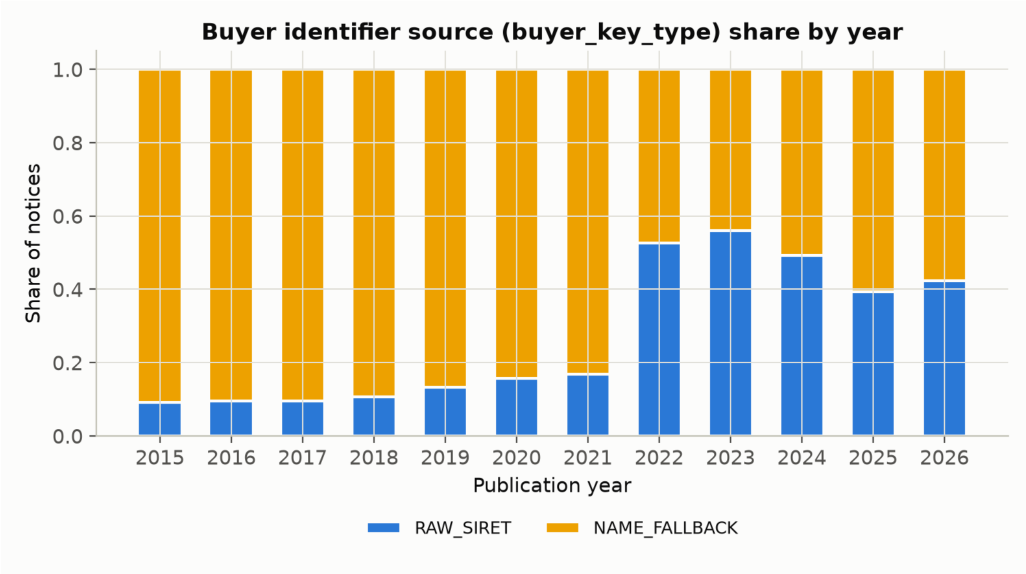

Figure 1: Buyer-identifier source (`buyer_key_type`) share by publication year, all 84,623 cleaned notices.

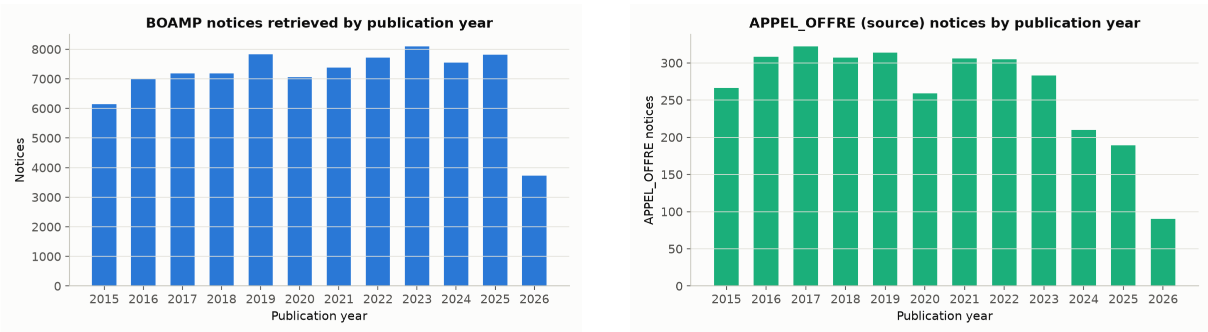

Figure 2: Left: all cleaned notices by publication year. Right: `APPEL_OFFRE` notices by publication year (all sectors, before the digital-scope filter).

4 Dataset Construction and Missing-Value Control

4.1 What Each Dataset Represents

BOAMP publishes notices, not longitudinal contract histories; a BOAMP row is one publication event. The pipeline therefore transforms a notice-level source into a contract-level survival dataset through six persisted stages (Table 6).

The 84,623-row cleaned table describes the available BOAMP evidence across *all* sectors and notice types; the 3,159-row source population describes possible digital-scope recurrence-source contracts; the survival handoff is the population on which every result in Sections 12–15 is measured.

Table 6: Official current data lineage (`reports/tables/audit_data_lineage.csv`, `audit_dataset_inventory.csv`).

Dataset	Unit	Rows	Cols	Creation script
`boamp_raw_flattened.csv`	notice	84,623	42	`parse_boamp.py`
`boamp_clean_m0_no_enrichment.csv`	notice	84,623	93	`preprocess_boamp_m0.py`
`boamp_m0_sources.csv`	source notice	3,159	22	`preprocess_boamp_m0.py`
`boamp_m0_candidate_pairs.csv`	source–candidate pair	6,137	19	`build_m0_candidate_pairs.py`
`boamp_m0_links_balanced.csv`	accepted link	618	21	`run_m0_linkage.py`
`boamp_survival_m0_balanced.csv`	survival unit	3,159	18	`run_m0_linkage.py`

4.2 How Missing Values Are Handled

The guiding rule, applied consistently across the pipeline, is **flag first, preserve raw values, impute only when a downstream step truly needs a value**. Table 7 documents the main decisions.

Table 7: Main missing-value and quality-control decisions.

Field	Observed quality	Handling rule
Buyer identity	27.2% of all notices carry a format-valid raw SIRET (99.6% of those also pass checksum).	Use `SIRET:...` when a valid 14-digit SIRET exists; else `NAME:...` from a normalized buyer name; `buyer_key_type` records which rule fired. No row is ever dropped for missing buyer identity — 0% `MISSING` across all 84,623 notices.
CPV code	78.4% of M0 sources have a cleaned 8-digit CPV code (down to 45–51% in 2024–2026, Section 5).	Derive the full hierarchy (`cpv_division/group/class/category`); flag division-only "generic" codes (9.15% of sources) separately; a missing CPV in a candidate pair is scored as a distinct `CPV_SCORE_MISSING`=0.1 (Section 7), never as an arbitrary neutral 0.5 and never as evidence of non-match.
Contract duration	Only 12.0% of M0 sources (378 of 3,159) have a directly observed, in-range declared duration; 88.0% required imputation.	Keep the raw declared value when in $[1, 120]$ months; else impute the median *observed* duration within the same CPV division (falling back to the global observed median). `dur_was_imputed` records this choice and is itself used as a survival covariate.
Contract start date	`ATTRIBUTION`-linked award dates cover 38.2% of eligible AO sources (1,207/3,159).	Use the linked `ATTRIBUTION` notice's publication date when `annonce_lie` resolves it; else fall back to the source's own `dateparution`. `start_date_source` records which rule fired.
Recurrence outcome	BOAMP has no explicit recurrence label.	Build `event` by the M0 linking rule (Section 7). If no accepted link is found before the study end date, `event`=0 and the row is right-censored, not deleted.

4.3 Why Right-Censoring Is Not Missing Data

The reference survival handoff contains 618 linked sources (`event`=1) and 2,541 unlinked-but-eligible sources (`event`=0), for a censoring rate of 80.4%. The unlinked rows are *not* removed: they still carry valid information — the source was observed, without a detected recurrence, for a known number of months. In survival-analysis notation:

- `event`=1: the M0 rule found and accepted a later same-buyer notice as a plausible recurrence; `time_to_event_or_censor_months` is the gap in months between the source and the accepted candidate's publication dates.
- `event`=0: no accepted candidate was found (either zero candidates passed blocking, or the best candidate's score fell below the variant's threshold); `time_to_event_or_censor_months` is the gap between the source's publication date and `study_end_date` (2026-07-13, the latest observed publication date in the corpus).

Removing censored rows would bias the model toward sources that recurred quickly and would erase long-running or late-window contracts from the risk set. An internal construction audit (`reports/tables/audit_survival_construction_checks.csv`) verified zero duplicated survival units, zero event rows with a missing linked candidate, zero censored rows carrying a candidate ID, zero nonpositive event/censoring times, and zero events recorded after the study end date.

5 Data Quality Issues Diagnosed

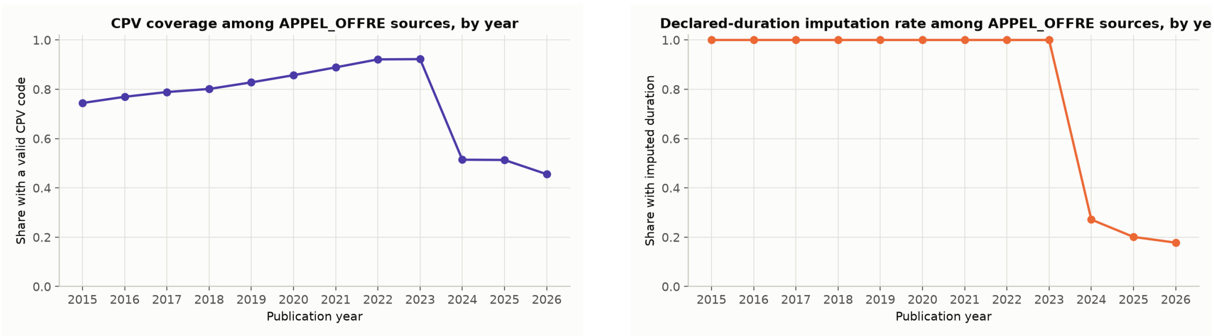

Figure 3: Left: CPV coverage among `APPEL_OFFRE` M0 sources, by year (item 3 above). Right: declared-duration imputation rate among M0 sources, by year (item 4 above).

6 Preprocessing Pipeline

6.1 Buyer Key Construction

For notice i with raw fields s_i (candidate SIRET), σ_i (candidate SIREN), and n_i (raw buyer name):

$$\texttt{buyer\_key}_i = \begin{cases} \texttt{SIRET:}s_i & \text{if } s_i \text{ passes format } \textit{and} \text{ Luhn checksum validation} \\ \texttt{SIREN:}\sigma_i & \text{else if a valid SIREN is present or derivable from a format-valid SIRET} \\ \texttt{NAME:}\phi(n_i) & \text{else, if } n_i \text{ is present} \\ \texttt{MISSING} & \text{otherwise} \end{cases} \tag{1}$$

where $\phi(\cdot)$ normalizes the buyer name (lowercase, strip accents, collapse whitespace). This is purely internal cleaning of BOAMP-provided values, never an external lookup. Across the

Table 8: Data quality issues and their impact on linking, per `reports/boamp_m0_preprocessing_report.md` §15 and `reports/tables/m0_preprocessing_quality_checks.csv`.

#	Issue	Observation	Impact on linking
1	Low SIRET coverage	Only 27.2% of all notices, 22.6% of M0 sources, carry a raw BOAMP SIRET.	77.4% of M0 sources fall back to a normalized-name buyer key.
2	Buyer name-variant fragmentation	Name-fallback buyers are matched only by exact normalized-name equality (no fuzzy matching implemented).	The same real buyer under two spellings splits into two `buyer_key` groups, producing structural false negatives that cannot be recovered by tuning the score.
3	CPV coverage declines in recent years	CPV coverage among M0 sources falls from ∼92% (2022–2023) to 45–51% (2024–2026), timed with the EU eForms/UBL schema transition (late 2023).	Recent-year `s_cpv` scores and generic-CPV shares should be read with this drop in mind (Figure 3).
4	Contract duration is sparsely observed	Only 378/3,159 (12.0%) M0 sources have a valid, non-imputed declared duration; median observed value is 6.0 months (n=378). Spot-checked across publication tiers (FNS 12.2%, JOUE 13.3%, MAPA 5.6% observed) — consistently low, not an eForms-specific artifact.	Drives 88.0% duration imputation, which in turn sets the temporal blocking window and the estimated-end-date-centered candidate search (Section 6).
5	Duration is a right-skewed, keyword-driven signal	Non-imputed durations cluster at 4 (25.9%), 48 (21.4%), 6 (11.9%), and 5 months (9.0%) rather than the smooth 12/24/36/48-month administrative ceilings a larger sample might show.	Because so few durations are directly observed, this distribution is itself low-confidence (n=378); see the imputation rule in Table 7.
6	Candidate reuse	In the balanced link set, 618 accepted links point to only 443 unique candidate notices; 121 candidates (27.3%) are reused by more than one source, max. multiplicity 6.	Prevents a naive one-to-one "renewal" interpretation without manual review (lots, framework agreements, or generic repeat buyers may be defensible many-to-one cases).
7	Blocking excludes most eligible sources	Only 1,236 of 3,159 eligible sources (39.1%) have ≥1 scored candidate.	No score-based diagnostic can recover a recurrence for the 1,923 zero-candidate sources; this is the single largest ceiling on the balanced event rate (Section 9).
8	Duration circularity	Declared/imputed duration defines the estimated end date, which centers both blocking and the temporal score, and duration also remains available as a downstream survival covariate.	A dedicated leakage audit (Section 15.3) shows the estimated duration effect on survival hazard is *not* stable to alternative, duration-independent linkage designs.
9	No verified ground truth	BOAMP provides no renewal field.	Precision/recall of the M0 rule are estimated only indirectly (Fellegi–Sunter mixture, Section 12.2) and remain unverified against manual labels (Section 16).

3,159 M0 sources: 713 (22.6%) resolve to `RAW_SIRET`, 0 to `RAW_SIREN` (no BOAMP notice in this scope carries a raw SIREN without also carrying a recoverable SIRET), 2,446 (77.4%) to `NAME_FALLBACK`, and 0 to `MISSING`.

6.2 Contract Start-Date Refinement

$$\texttt{start\_date}_i = \begin{cases} \texttt{publication\_date}(j) & \text{if } \exists\, j \in \texttt{ATTRIBUTION} : \texttt{annonce\_lie}(j) \ni i, \text{ earliest such } j \\ \texttt{publication\_date}_i & \text{otherwise} \end{cases} \tag{2}$$

In practice, 1,207 of 3,159 M0 sources (38.2%) obtain a more precise start date via a linked `ATTRIBUTION` notice; the remaining 61.8% fall back to their own publication date.

6.3 Duration Imputation

$$d_i = \begin{cases} \texttt{declared\_duration\_months}_i & \text{if observed and} \in [1, 120] \text{ months} \\ \text{median}\{d_k : k \in \texttt{cpv\_division}(i), \text{ observed}\} & \text{else, if that median exists} \\ \text{median}\{d_k : \text{observed}, k \in \text{digital-scope AO}\} & \text{else (global fallback)} \end{cases} \tag{3}$$

Medians are computed on the *digital-scope, Pays-de-la-Loire* ***APPEL_OFFRE*** population only (i.e. the 3,159-row M0 source population), so imputed durations reflect in-scope contracts rather than importing behavior from other sectors. 2,781 of 3,159 sources (88.0%) receive an imputed value (`dur_was_imputed=True`).

6.4 Estimated End Date

$$\hat{e}_i = \texttt{start\_date}_i + d_i \text{ months} \tag{4}$$

This is the center of both the temporal-blocking window (Section 7) and the survival-relevant "expected recurrence date."

6.5 CPV Cleaning and Hierarchy

A raw CPV code is kept only if it matches an 8-digit pattern; from it, `cpv_division` (2 digits), `cpv_group` (3), `cpv_class` (4), and `cpv_category` (5) are derived by left truncation. A code ending in six zeros (`XX000000`) is flagged `cpv_generic_flag = True`, since it carries no subdivision signal.

6.6 Text Normalization

The free-text `objet` field is cleaned (`clean_objet`: strip boilerplate, normalize whitespace/accents) into `objet_clean`, which feeds a TF-IDF vectorizer (`scikit-learn`, unigrams+bigrams, `max_features=50,000`, `min_df=2`) fitted over all 3,159 eligible source texts. Sentence-Transformer embeddings were evaluated for this step but not adopted: at the pipeline's original national scale (∼340k notices) CPU-only transformer embedding was judged impractical, and TF-IDF+cosine was kept for continuity even though the current Pays-de-la-Loire/digital scope (3,159 sources) is small enough that embeddings would now be tractable — listed as a next step in `reports/boamp_m0_preprocessing_report.md` §16.

7 Recurrence-Linking Algorithm (M0)

7.1 Position in the Record-Linkage Literature

Methodologically, M0 is a *record-linkage* problem in the classical sense of Fellegi and Sunter [1]: deciding, from a vector of field-comparison scores, whether two records refer to the same underlying entity — here, the same recurring procurement need rather than the same person or firm. The two standard ingredients both appear in M0 in deterministic form: *blocking* (restricting comparisons to pairs sharing a cheap, high-precision key — here the buyer key plus a temporal window [2]) and *comparison-vector scoring* (here a fixed linear combination rather than estimated log-likelihood ratios). What is project-specific is (i) the target relation itself — recurrence of a need, which has no ground-truth registry to link against, unlike classical deduplication — and (ii) the deterministic, fully documented weights, chosen for auditability over statistical optimality. The probabilistic Fellegi–Sunter machinery is used in this project only *diagnostically*, as an unsupervised mixture re-analysis of M0's comparison vectors (Section 12.2), not as the linkage rule.

7.2 Notation

Let i index an eligible source notice (non-`MISSING` buyer key) and j a later candidate notice from the *same* buyer key. Each notice carries a publication date t, an estimated end date $\hat{e}$ (for sources), a CPV hierarchy, cleaned object text, and a buyer-key type.

7.3 Step 1 — Eligibility and Candidate Pool

The temporal window is derived from data, not hardcoded:

$$W = \mathrm{clip}\big(\mathrm{round}(0.5 \times \tilde{d}_{\mathrm{obs}}),\ 6,\ 24\big) \text{ months} \tag{5}$$

where $\tilde{d}_{\mathrm{obs}} = 6.0$ months is the median *observed* (non-imputed) duration among M0 sources (Section 5, item 4). This gives $W = \mathrm{clip}(3, 6, 24) = 6$ months. The realized candidate universe is

$$\Omega = \big\{(i,j) : \texttt{buyer\_key}(j) = \texttt{buyer\_key}(i),\ t_j > t_i,\ |t_j - \hat{e}_i| \leq W\big\}, \tag{6}$$

capped, per source, at the 30 temporally nearest candidates (`MAX_CANDIDATES_PER_SOURCE`) for runtime; this cap did not bind in the current (much smaller than the pipeline's original national) scope. Ω contains 6,137 pairs covering 1,236 of the 3,159 eligible sources (39.1% blocking coverage, Section 9).

7.4 Step 2 — Component Scores

Text similarity is TF-IDF [3] cosine similarity between source and candidate object text:

$$s_{\mathrm{text}}(i,j) = \frac{\mathbf{v}_i \cdot \mathbf{v}_j}{\|\mathbf{v}_i\|\|\mathbf{v}_j\|} \in [0,1]. \tag{7}$$

CPV compatibility follows the official CPV hierarchy (division / group / class / category / full 8-digit code), rewarding the depth of the longest shared prefix:

$$s_{\mathrm{cpv}}(i,j) = \begin{cases} 0.1 & \text{either CPV missing} \\ 1.0 & \text{identical full 8-digit code} \\ 0.8 & \text{same 5-digit category} \\ 0.6 & \text{same 4-digit class} \\ 0.4 & \text{same 3-digit group} \\ 0.2 & \text{same 2-digit division} \\ 0.0 & \text{valid CPVs, different divisions} \end{cases} \tag{8}$$

A missing CPV is scored 0.1 — a documented weak-neutral value, distinct from and strictly lower than any confirmed match, but not treated as evidence *against* the pair the way a confirmed cross-division mismatch (0.0) is.

Temporal proximity decays linearly from 1.0 at the estimated end date to 0.0 at the window boundary:

$$s_{\text{time}}(i,j) = \text{clip}\left(1 - \frac{|(t_j - t_i) - d_i|}{W},\ 0,\ 1\right). \tag{9}$$

Buyer-key reliability reflects how trustworthy the already-enforced buyer-key match is, not whether it matched (candidate generation already conditions on identical `buyer_key`):

$$s_{\text{buyer}}(i) = \begin{cases} 1.00 & \texttt{buyer\_key\_type} = \texttt{RAW\_SIRET} \\ 0.85 & \texttt{buyer\_key\_type} = \texttt{RAW\_SIREN} \\ 0.60 & \texttt{buyer\_key\_type} = \texttt{NAME\_FALLBACK} \end{cases} \tag{10}$$

7.5 Step 3 — Composite Score

$$S(i,j) = 0.35\, s_{\text{text}} + 0.30\, s_{\text{cpv}} + 0.25\, s_{\text{time}} + 0.10\, s_{\text{buyer}} \in [0,1]. \tag{11}$$

Weights are documented, fixed choices, not fitted: text and CPV are treated as the primary content-based recurrence signals, timing anchors the match around the expected end date, and the buyer-reliability term mostly discounts low-confidence name-only matches without changing the ranking *within* a source's candidate pool (since it is constant across all of that source's candidates). Weights were reviewed against the observed score distribution (mean composite score 0.250 over all pairs) but not re-tuned against a validation set, to avoid overfitting to a single run's distribution.

7.6 Step 4 — Best-Match Selection and Threshold Variants

For each source with ≥ 1 candidate, candidates are ranked by $S(i,j)$ and the rank-1 (best) candidate's score is kept. Three link *variants* are derived from the 25th/50th/75th percentile of the rank-1 score distribution ($n = 1{,}236$), not from an a-priori cutoff:

Table 9: M0 link variants (`reports/tables/m0_method_summary.csv`). Denominator is the full eligible population, 3,159 sources.

Variant	Threshold τ	Linked events	Event rate	Median gap
Broad	0.2642 (p25)	927	29.3%	37.9 months
Balanced (reference)	**0.3230 (p50)**	**618**	**19.6%**	**37.7 months**
Strict	0.3931 (p75)	309	9.8%	44.7 months

Because all three thresholds are cuts on the *same* rank-1 score distribution, the sets are nested: `strict` ⊂ `balanced` ⊂ `broad`. The reference outcome variable is

$$\texttt{event}_i = \begin{cases} 1 & \text{if } \exists j : S(i,j) = \max_k S(i,k) \geq \tau_{\text{balanced}} \\ 0 & \text{otherwise} \end{cases} \tag{12}$$

`event`=0 means *no accepted candidate under the current rule*, not verified evidence that no recurrence occurred.

8 Output Dataset — Survival Handoff

8.1 File: `boamp_survival_m0_balanced.csv`

One row per eligible source notice (3,159 rows, 18 columns); the official Phase-2 modeling input.

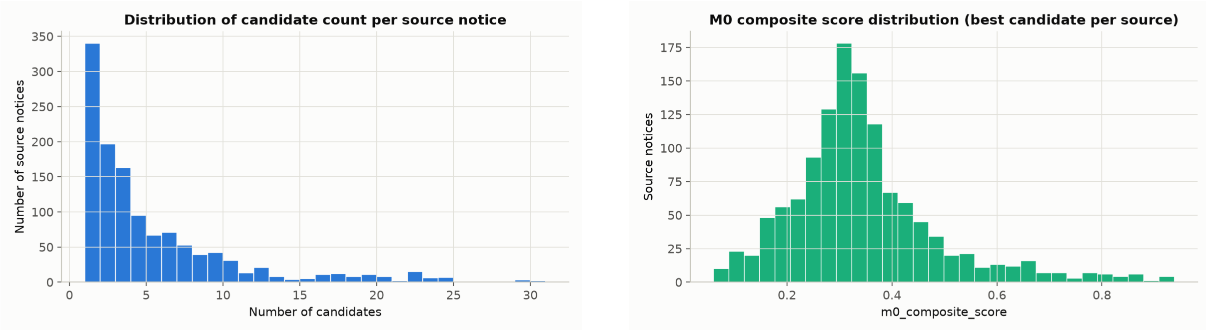

Figure 4: Left: distribution of the number of scored candidates per source notice, among the 1,236 sources with ≥ 1 candidate. Right: distribution of the best (rank-1) composite score per source, with the broad/balanced/strict thresholds marked.

Table 10: Column dictionary — `boamp_survival_m0_balanced.csv`.

Column	Type	Description
`notice_id`	string	Source `APPEL_OFFRE` BOAMP ID.
`buyer_key` / `buyer_key_type`	string	Grouping key and its provenance (RAW_SIRET / NAME_FALLBACK).
`publication_date`, `start_date`, `estimated_end_date`, `study_end_date`	date	Temporal anchors (Section 6).
`cpv_division`, `cpv_category`, `category_label`	string	Classification of the source contract; `category_label` is binary (`DIGITAL_ICT` for every row here, since the file is already scope-filtered).
`declared_duration_months`	float	Observed or imputed duration (Section 6).
`dur_was_imputed`	bool	True for 2,781/3,159 rows (88.0%).
`event`	int	**Target variable.** 1 = accepted M0 link (618 rows), 0 = no accepted link (2,541 rows).
`time_to_event_or_censor_months`	float	Survival time, always filled. For `event`=1: gap in months between source and linked candidate publication dates. For `event`=0: gap between source publication date and `study_end_date`.
`linked_candidate_notice_id`	string	Accepted candidate's BOAMP ID; empty when `event`=0.
`m0_composite_score`	float	$S(i, j^*)$ for linked rows; empty when `event`=0.
`variant`	string	Constant `"balanced"` in this file.

8.2 Score Distributions for Linked Events

Table 11 compares component-score statistics across five populations: all 6,137 candidate pairs, the 1,236 rank-1 (best) candidates, the 618 accepted balanced links, the 5,519 rejected candidates, and the 896 runner-up candidates (rank-2 among sources with ≥ 2 candidates). This is a placebo-style comparison: if the composite score carried no discriminative signal, accepted links and runner-ups would have similar distributions.

The mean composite score of accepted links (0.444) is well above the rejected-candidate mean (0.229), and the margin m (best minus second-best score) is also larger for accepted links (0.161) than for the ambient candidate pool (0.086), consistent with the score being internally discriminative (Section 9.3 quantifies this further).

Table 11: Composite score by population (source: `reports/tables/audit_score_component_distributions.csv`).

Population	n	Mean S	Median S	Mean margin m
All candidate pairs	6,137	0.250	0.241	0.086
Rank-1 (best) candidates	1,236	0.345	0.323	0.137
Accepted balanced links	**618**	**0.444**	**0.393**	**0.161**
Rejected candidates	5,519	0.229	0.226	0.077
Runner-up candidates	896	0.287	0.289	0.080

9 Linking Diagnostics

9.1 Blocking Coverage

Of 3,159 eligible sources, only 1,236 (39.1%) have ≥ 1 scored candidate; 1,923 (60.9%) have zero. No downstream score-based diagnostic — Fellegi–Sunter posteriors, threshold sensitivity, corruption recovery — can recover a recurrence signal for these 1,923 sources, since they never enter Ω. Coverage decays sharply toward the end of the study period (Table 12), the expected signature of finite follow-up combined with duration-centered blocking.

Table 12: Blocking coverage by publication year (`reports/tables/audit_blocking_coverage.csv`).

Year	Eligible sources	With candidate	Zero candidate	Coverage
2015	266	151	115	56.8%
2016	308	177	131	57.5%
2017	322	191	131	59.3%
2018	307	156	151	50.8%
2019	314	165	149	52.5%
2020	259	109	150	42.1%
2021	306	95	211	31.0%
2022	305	60	245	19.7%
2023	283	18	265	6.4%
2024	210	50	160	23.8%
2025	189	53	136	28.0%
2026 (partial)	90	11	79	12.2%

Coverage also varies by buyer publication frequency: 0% for buyers with a single digital AO notice in the whole period (386 sources, structurally excluded), rising to 69.0% for buyers with 26+ digital AO notices (963 sources). `RAW_SIRET`-keyed sources have *lower* coverage (29.3%) than `NAME_FALLBACK`-keyed sources (42.0%) — a reminder that buyer-key reliability and blocking coverage are not the same axis of credibility.

9.2 Zero-Candidate Failure Analysis

The audit table `audit_censoring_diagnostics.csv` decomposes the 1,923 zero-candidate sources into two structurally distinct causes, distinguished from the 1,236 sources that *do* have candidates:

The temporal-window-excluded cause (1,125 sources, 35.6% of all eligible sources) is the largest single failure mode and is directly tied to the duration-centered blocking design audited

Table 13: Zero-candidate failure decomposition (all 3,159 eligible sources).

Cause	Count	Interpretation
Has ≥ 1 candidate	1,236	Enters Ω; may or may not clear the balanced threshold.
No later same-buyer notice at all	798	Structural: this buyer published no further digital AO in the study period.
Later notices exist, but all fall outside the ±6-month window	1,125	Structural: recurrence timing (if any) is outside the duration-implied search window.

in Section 15.3; Section 10 gives a concrete worked example of this failure mode.

9.3 Score Margin and Threshold Sensitivity

The score margin m (best minus second-best composite score) is defined for all 1,236 rank-1 candidates: the rank-1-minus-rank-2 score when a source has ≥ 2 candidates, or the rank-1 score itself when only one candidate exists. The mean margin among accepted balanced links is 0.161 (Table 11), versus 0.086 across all pairs.

Perturbing the balanced threshold by $\pm\delta$ changes how many rank-1 sources flip event status:

Table 14: Threshold perturbation sensitivity (`reports/tables/m6a_threshold_sensitivity.csv`), around $\tau_{\text{balanced}} = 0.3230$.

$\pm\delta$	Sources flipped	Share of 1,236 rank-1 sources
0.005	48	3.9%
0.010	88	7.1%
0.020	227	18.4%
0.050	516	41.7%

The balanced threshold sits in a non-negligible mass of the rank-1 score distribution: borderline cases are common (Figure 5) and should not be presented as individually reliable without manual or external corroboration.

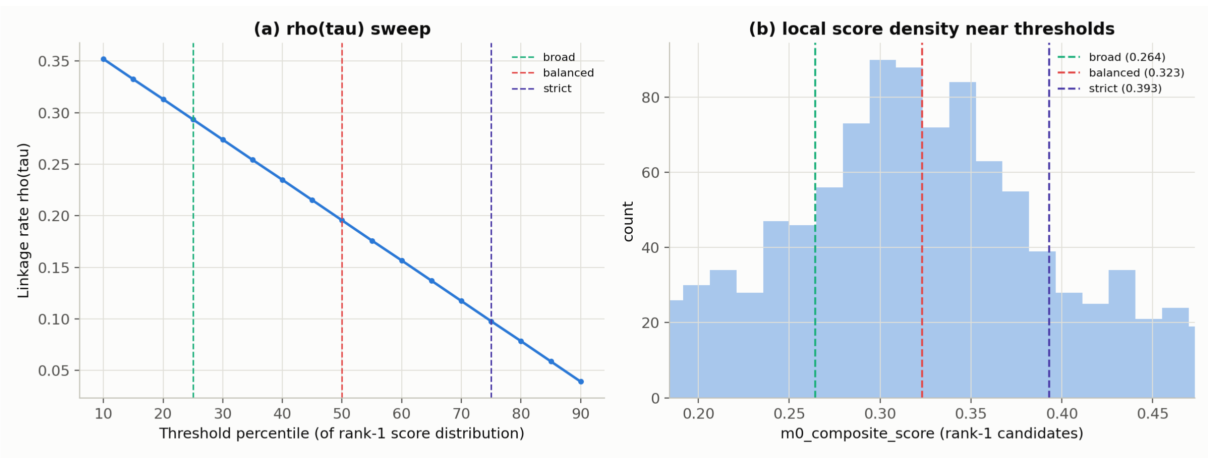

Figure 5: (a) Linkage rate $\rho(\tau)$ as the threshold percentile sweeps from p10 to p90 of the rank-1 score distribution. (b) Local density of rank-1 scores near the three named thresholds.

9.4 Feature Ablation

Dropping one score component, renormalizing the remaining weights, reranking candidates, and rebuilding a same-sized link set gives the Jaccard overlap with the reference balanced set shown in Figure 6. The buyer component is least influential (Jaccard 0.904) because s_{buyer} is constant across all candidates for a given source — it can move a source above or below threshold but cannot change which candidate ranks first *within* that source. Time (Jaccard 0.668) and text (0.684) are the most influential despite text's larger nominal weight, because both vary continuously within a candidate pool while CPV is coarser (only 6 discrete rungs).

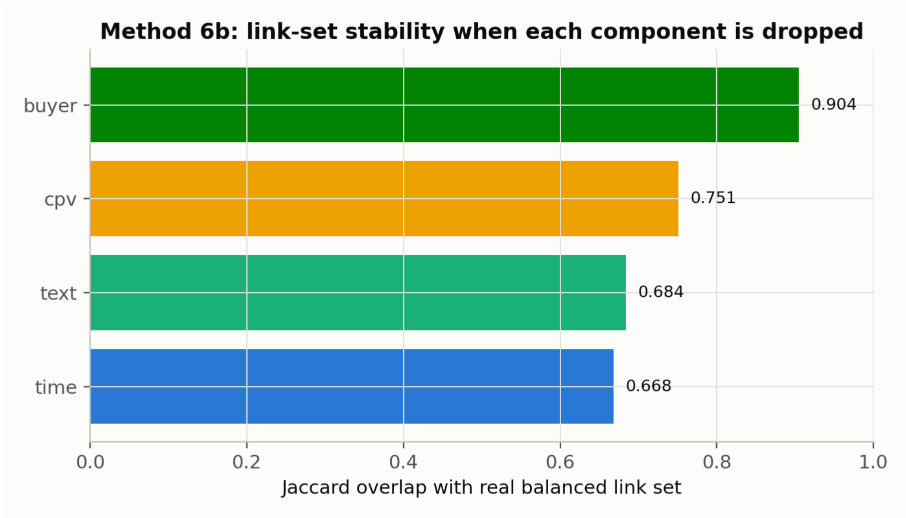

Figure 6: Link-set stability (Jaccard overlap with the reference balanced set) when each score component is dropped and weights renormalized.

9.5 Linkage Rate by Publication Year

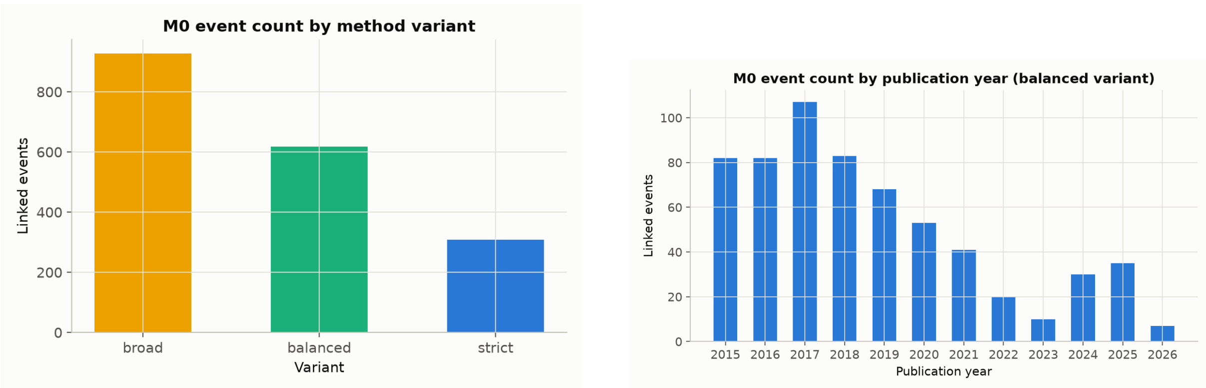

Figure 7: Left: event count by variant (broad/balanced/strict). Right: balanced-variant event count by publication year — coverage decay (Table 12) directly shrinks the number of events available in recent cohorts.

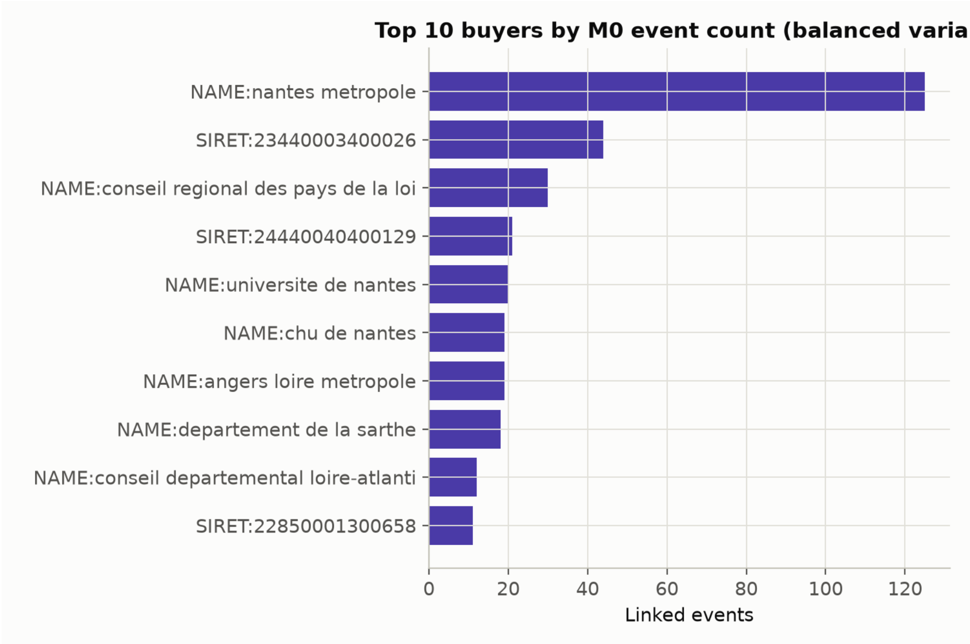

Figure 8: Top 10 buyers by balanced-variant event count. The top buyer, `NAME:nantes metropole`, contributes 125 of 618 events (20.2%) — notably a name-keyed, not SIRET-keyed, buyer.

10 Worked Examples

10.1 Worked Link Example

Source notice `15-139094` (SDIS de la Vendée, buyer key `NAME:sdis 85`) was published 2015-09-10, object "*Acquisition d'appareils respiratoires isolants à circuit ouvert et de bouteilles d'air pour le SDIS de la Vendée à La Roche-sur-Yon*" (self-contained breathing apparatus), CPV `35111100` (division 35, class 3511, category 35111). Its declared duration was missing and was imputed to 48 months (the division-35 observed median), giving $\hat{e} = 2015\text{-}12\text{-}07 + 48$ mo $= 2019\text{-}12\text{-}07$.

Three same-buyer candidates fell inside the ±6-month window around $\hat{e}$; their full scored rows, exactly as stored in `boamp_m0_candidate_pairs.csv`, are:

Table 15: Complete candidate pool for source `15-139094` (`data/processed/boamp_m0_candidate_pairs.csv`).

Candidate	Gap (mo)	s_{time}	s_{text}	s_{cpv}	s_{buyer}	$S(i,j)$	Rank
`19-141668`	48.29	0.5675	0.1336	0.6	0.6	**0.4286**	1
`20-72017`	56.80	0.0145	0.3058	0.6	0.6	0.3507	2
`19-108347`	46.02	0.1897	0.0090	0.0	0.6	0.1106	3

The best-scoring candidate is `19-141668`, published 2019-09-19, object "*Acquisition d'un simulateur de feux réels avec traitement des fumées destiné à la formation des sapeurs-pompiers du SDIS de la Vendée*" (a fire-training simulator), CPV `35110000` (division 35, class 3511, category 35110). Note how the pool illustrates the scoring trade-offs: the rank-2 candidate actually has *higher* text similarity (0.306 vs. 0.134) but lands at the far edge of the temporal window ($s_{\text{time}} = 0.014$), while the rank-3 candidate is temporally closer but nearly text-orthogonal and in a different CPV division ($s_{\text{cpv}} = 0$).

This pair had a margin of 0.078 over the second-best of 3 candidates and was selected as source `15-139094`'s best match; because $0.4286 \geq \tau_{\text{balanced}} = 0.3230$ (and also $\geq \tau_{\text{strict}} =$

Table 16: Score calculation for the pair (`15-139094` → `19-141668`).

Quantity	Formula / value	Result
$t_j - t_i$	gap in months	48.29
$\lvert(t_j - t_i) - d_i\rvert$	$\lvert 48.29 - 48\rvert$	0.29
s_{time}	$1 - 0.29/6$	0.5675
s_{text}	TF-IDF cosine of the two objects	0.1336
s_{cpv}	same 4-digit class (3511), different 5-digit category	0.6000
s_{buyer}	`NAME_FALLBACK`	0.6000
$S(i, j)$	$0.35(0.1336) + 0.30(0.6) + 0.25(0.5675) + 0.10(0.6)$	**0.4286**

0.3931), the source is recorded as `event`=1 with `time_to_event_or_censor_months`=48.29 in `boamp_survival_m0_balanced.csv`. Downstream, this row contributes one event at $t = 48.29$ months to the balanced Kaplan–Meier curve of Figure 12, one uncensored observation to the division-35 stratum in Figure 13, and one `dur_was_imputed`=`True` event to the Cox model of Section 14.3 — and, because its 48-month duration was itself imputed, it is also a concrete instance of the duration-circularity mechanism audited in Section 15.3: the imputed duration placed the search window that found the candidate that became the event. This example is instructive precisely because the text score is low (0.134, well below the accepted-link mean of 0.305): the link survives on the combination of CPV-class agreement and near-exact temporal alignment with the (imputed) expected end date, illustrating why the composite design — rather than a single dominant signal — matters, but also why low-text-similarity links deserve scrutiny in manual review (Section 16).

10.2 Worked Censoring Example

Source notice `24-10209` (SEMITAN, Nantes's public transport operator, buyer key `NAME:semitan`) was published 2024-01-29, object "*Renouvellement des infrastructures Serveurs - Licences Vmware*" (server infrastructure and VMware license renewal), CPV `72212218` (division 72, IT services). Its declared duration was *directly observed* (not imputed) at 36 months, giving $\hat{e} = 2027\text{-}01\text{-}29$.

SEMITAN is a frequent buyer with 32 digital AO sources across the study period, so this is not a case of "no later same-buyer notice" — indeed six further SEMITAN notices were published in 2024–2025 alone. But the ±6-month candidate window around $\hat{e}$ spans only 2026-07-29–2027-07-29, and the latest SEMITAN notice observed in the corpus (`25-97524`, 2025-09-02) still falls nearly a year before that window opens. The source therefore has *zero* candidates — the "temporal-window-excluded" failure mode of Section 9 — and is recorded as `event`=0 with `time_to_event_or_censor_months` = (2026-07-13 − 2024-01-29) ≈ 29.4 months. In survival language, the only valid statement is "no recurrence was detected by month 29.4 under the current ±6-month window"; it is neither evidence that SEMITAN will not re-tender this need, nor evidence that it will.

11 Limitations and Risks

L1. No ground truth. BOAMP provides no renewal field; the M0 event is entirely a constructed proxy. Precision/recall are only estimated indirectly (Section 12.2) and not yet validated against manual labels (Section 16).

L2. Blocking, not scoring, is the binding constraint. 60.9% of eligible sources never enter the candidate universe Ω (Section 9.1); this ceiling cannot be moved by tuning score weights or thresholds.

L3. Duration circularity. Declared/imputed duration constructs the estimated end date used for blocking *and* temporal scoring, and duration remains available downstream as a survival covariate. The estimated duration hazard-ratio sign is not stable to alternative, duration-independent linkage designs (Section 15.3).

L4. Name-fallback buyer fragmentation. 77.4% of M0 sources are name-keyed; no fuzzy-name or external-registry reconciliation is implemented, so buyer-name variants split into separate groups and produce structural false negatives.

L5. Candidate reuse. 27.3% of accepted balanced-link candidates are claimed as the best match by more than one source (max. multiplicity 6), which is not automatically an error (lots, frameworks) but prevents a naive one-to-one interpretation without manual review.

L6. Sample scope. The dataset covers only Pays de la Loire, digital/ICT contracts, 2015–2026. Linking rates, score distributions, and CPV-division rankings may differ substantially elsewhere.

L7. Sparse survival covariates. The Cox model's CPV-division dummy structure produces convergence/separation warnings for rare divisions (Section 14.3), and temporal (train/test) validation was attempted but deferred as unstable (Section 15).

12 Dataset Reliability Checks

12.1 Confidence Tiers

Because `strict` $\subset$ `balanced` $\subset$ `broad` (Section 7), the three named thresholds naturally partition the 927 broad-linked sources into three confidence tiers of near-identical size (each spans a 25-percentile-point band of the rank-1 score distribution):

Table 17: Confidence tiers derived from the nested broad/balanced/strict thresholds.

Tier	Count	Composite-score band
HIGH (= strict set)	309	$S \geq 0.3931$
MEDIUM (balanced minus strict)	309	$0.3230 \leq S < 0.3931$
LOW (broad minus balanced)	309	$0.2642 \leq S < 0.3230$
Not linked even at broad threshold	927 (of 1,236 blocked)	$S < 0.2642$
Blocked, zero candidates	1,923	N/A

The reference (balanced) event set is HIGH $\cup$ MEDIUM (618 sources). Restricting further to HIGH-only (strict, 309 events) is the conservative sensitivity check used throughout Section 15.

12.2 Unsupervised Fellegi–Sunter Latent-Mixture Diagnostic

Following the latent-class reading of the Fellegi–Sunter model [1], in which match (M) and non-match (U) parameters can be estimated without labeled data via the EM algorithm [4, 5], an unsupervised two-class Beta mixture (match class M, non-match class U) was fitted by EM over the comparison vectors $\gamma_j = (s_{\text{text}}, s_{\text{cpv}}, s_{\text{time}}, s_{\text{buyer}})$ of *all* 6,137 pairs in Ω, not only rank-1 winners (Figure 9). This is a model-based internal diagnostic, not external validation: `s_cpv` is discrete and `s_buyer` is nearly source-level rather than pair-level, both approximated by continuous Beta densities. EM converged to log-likelihood 40,975.5 (identical across warm-start and 6 total restarts, Jaccard=1.0 on posteriors > 0.5), estimated mixing weight $\hat{p} = 0.181$.

For posterior match probability $r_j = P(M \mid \gamma_j)$:

$$\widehat{\text{precision}}_{\text{FS}} = \frac{\sum_{j \in L_{\text{balanced}}} r_j}{|L_{\text{balanced}}|} = 0.455, \qquad \widehat{\text{recall}}_{\Omega} = \frac{\sum_{j \in L_{\text{balanced}}} r_j}{\sum_{j \in \Omega} r_j} = 0.253. \tag{13}$$

$\widehat{\text{recall}}_{\Omega}$ is conditional on Ω and explicitly excludes the 1,923 zero-candidate sources; it cannot be read as recall over all possible true recurrences. A conditional-independence check between the two continuous components found weak correlation (Pearson $r = 0.047$ between `s_cpv` and `s_text` on the 618 linked pairs, Spearman $r = -0.106$, both below the 0.3 flag threshold), supporting — but not proving — the model's conditional-independence assumption.

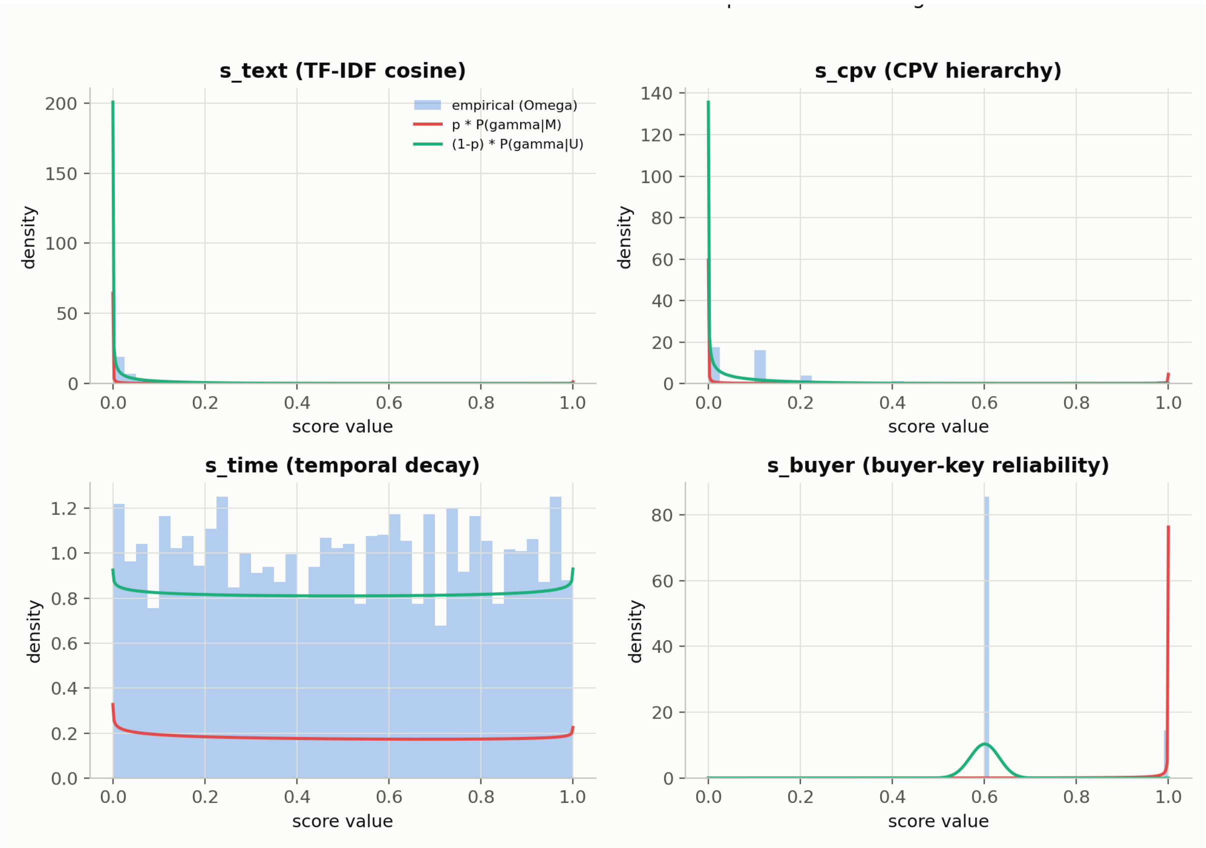

Figure 9: Fitted Beta-mixture densities ($p \cdot P(\gamma \mid M)$ in red, $(1 - p) \cdot P(\gamma \mid U)$ in aqua) against empirical score histograms, for each of the four score components over Ω ($n = 6{,}137$ pairs).

12.3 Semi-Synthetic Corruption Recovery

Using the 309 strict links as a conservative (not ground-truth) reference subset, text/CPV/duration fields were corrupted at increasing severity levels 0–3; a pair is "recovered" if the true candidate remains top-ranked in its (fixed) real candidate pool and still clears the balanced threshold.

Text degradation is the main breaking point; duration-only corruption is unusually robust because most strict-link sources already carry imputed durations close to their CPV-division median, so perturbing an already-imputed value changes little. This experiment holds the candidate pool fixed and so does *not* test corruption effects that would alter blocking itself — a full reblocking corruption study remains deferred (Section 16).

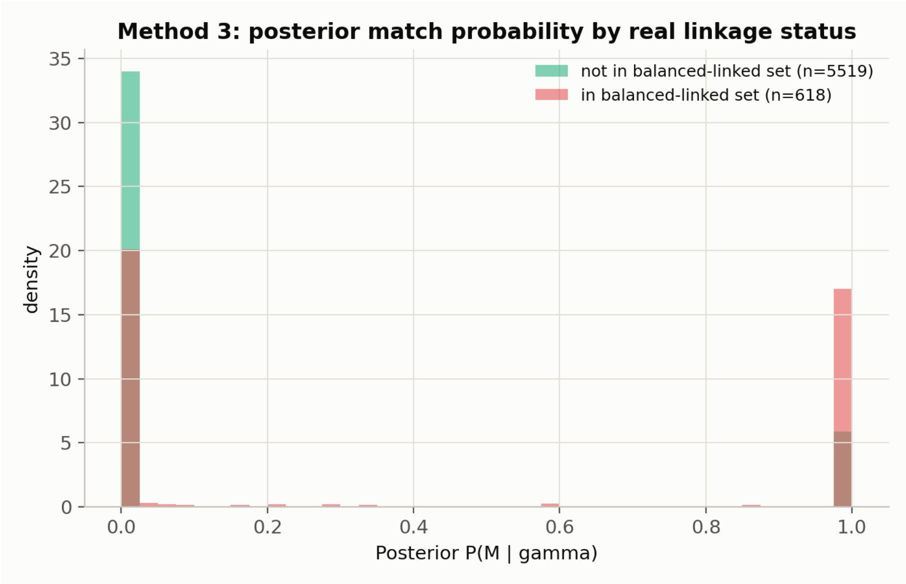

Figure 10: Posterior match probability $P(M \mid \gamma)$, split by whether the pair is in the balanced-linked set. The two distributions separate but overlap substantially, consistent with the moderate model-based precision estimate above.

Table 18: Recovery rate R(severity) under semi-synthetic corruption (`reports/tables/m4_recovery_summary.csv`), $n = 309$.

Sweep	R_0	R_1	R_2	R_3	$R_0 - R_3$
Combined	1.000	0.906	0.654	0.524	0.476
Text only	1.000	0.916	0.777	0.718	0.282
CPV only	1.000	0.990	0.984	0.851	0.149
Duration only	1.000	0.987	0.864	0.981	0.019

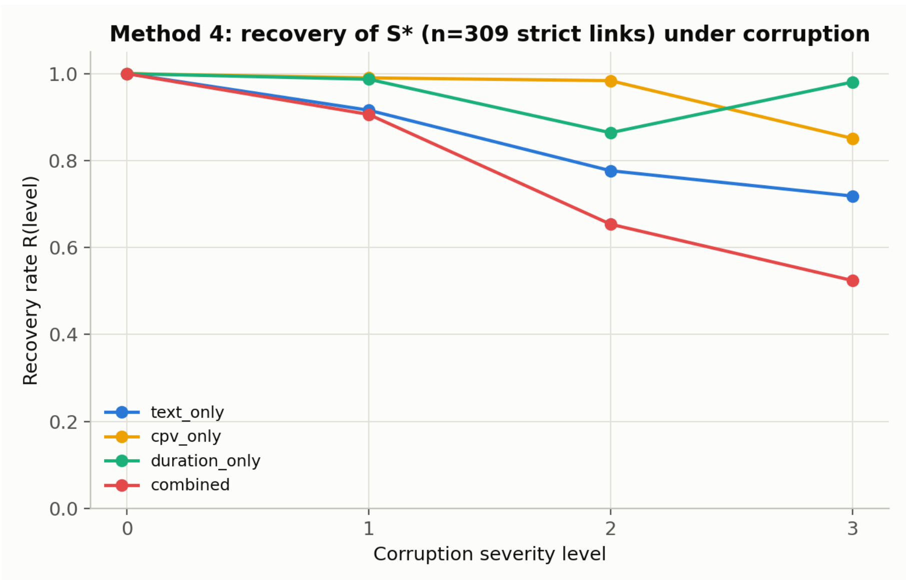

Figure 11: Recovery rate of the true strict-link candidate under increasing per-field corruption severity, by which field is corrupted.

12.4 Statistical Roadmap

Table 19 separates the different roles these tools play, following the credibility-audit discipline that a statistically significant result is not the same as a good ranking metric, and a stable internal diagnostic is not external validation.

Table 19: Statistical roadmap for this report.

Tool	Type	Question answered	Main limitation
Fellegi–Sunter mixture	Model-based estimator	Model-based precision/recall on Ω?	Conditional on blocking; not verified accuracy.
Corruption recovery	Semi-synthetic diagnostic	Robustness of strict links to field degradation?	Fixed candidate pool; does not test reblocking.
Threshold sensitivity	Sensitivity sweep	How many links flip near τ_{balanced}?	Describes ambiguity, not correctness.
Feature ablation	Sensitivity sweep	Which component drives the ranking?	Score-internal; not validated against labels.
Kaplan–Meier	Estimator	Time-to-linked-recurrence distribution?	Estimates the proxy event's timing, not legal renewal.
Cox PH model	Regression	Which covariates associate with the proxy hazard?	Associations conditional on the constructed event; not causal.
Duration-leakage audit	Design-sensitivity check	Is the duration effect robust to linkage design?	Answer here is *no* (Section 15.3).

13 Survival Analysis: Methods

13.1 Dataset Overview

Table 20: Reference survival dataset summary (`boamp_survival_m0_balanced.csv`).

Metric	Value
Total sources (rows)	3,159
Events (`event`=1)	618 (19.6%)
Censored (`event`=0)	2,541 (80.4%)
Study period	2015-03-02 to 2026-07-13
Median linked gap (events only)	37.7 months

13.2 Notation and Methods

Let $T_i^{\text{obs}} = \min(T_i, C_i)$ and $\delta_i = \mathbf{1}[T_i \leq C_i]$, stored as `time_to_event_or_censor_months` and `event`.

Kaplan–Meier estimator [6]. $\hat{S}(t) = \prod_{j:t_j \leq t}(1 - d_j/n_j)$, the probability of not-yet-linked survival past t, where n_j is the number at risk just before event time t_j and d_j the number of events at t_j. Its validity as an unbiased estimator requires censoring to be non-informative — an assumption that is *not* guaranteed here, since censoring mixes administrative end-of-study with linkage failure (blocking exclusion, sub-threshold scores); the censoring-balance diagnostic in `audit_censoring_diagnostics.csv` shows censored sources differ systematically from event sources (e.g. far fewer scored candidates and lower buyer publication frequency), so KM curves must be read as descriptive summaries of the constructed proxy process, not as unbiased estimates of a well-defined recurrence-time distribution.

Restricted mean survival time (RMST) at horizon $\tau = 60$ months, $\int_0^\tau \hat{S}(t)\,dt$ — the model-free "average event-free months out of the first 60" summary recommended when the median is not reached [9] — is used as the primary summary statistic *instead of* the median whenever $\hat{S}(t)$ never crosses 0.5 within the observed range (which is the case for every variant in this study, Section 14).

Cox proportional hazards model [7]: $h(t \mid \mathbf{x}_i) = h_0(t)\exp(\boldsymbol{\beta}^\top \mathbf{x}_i)$, fit with `lifelines` [11] (penalizer = 0.01), with covariates log(1 + declared duration months), the duration-imputation flag, CPV-division dummies, and buyer-key-type dummies, and robust standard errors clustered by `buyer_key` (a source contract and its own linked candidate never share a cluster, but two different sources from the same buyer do, and buyer-level clustering accounts for that dependence). Hazard ratio $\mathrm{HR}_k = \exp(\beta_k)$; > 1 means a higher instantaneous linking rate (faster proxy recurrence).

Proportional-hazards diagnostic: scaled-Schoenfeld-residual test (H_0: PH holds) [8] via `lifelines`' `check_assumptions`.

Parametric AFT models: Weibull and log-normal accelerated failure-time models, same covariate set, compared by AIC (lower is a better fit/complexity trade-off among models on the same data) and in-sample concordance.

Functional form check: a quadratic term $(\log(1+d))^2$ added to the clustered Cox model, to test whether the duration effect is linear on the log-hazard scale.

Prediction calibration: for horizons $h \in \{12, 24\}$ months, predicted risk $1 - \hat{S}(h \mid \mathbf{x}_i)$ from the fitted Cox model is grouped into quintiles and compared against the observed $\mathbf{1}[\texttt{event} = 1 \wedge T^{\mathrm{obs}} \leq h]$ rate and Brier score [10] per quintile.

All of the above are refit on eight event definitions (reference balanced, broad, strict, and five duration/window sensitivity designs) to test robustness (Section 15).

14 Survival Results

14.1 Kaplan–Meier Survival Estimates

Figure 12 shows the overall KM curve for all three named variants. **None of the three curves cross 50% survival**: median survival is undefined (`inf`) for broad, balanced, and strict alike, so RMST at 60 months is the primary descriptive summary (Table 21).

Table 21: Overall survival summary by variant (`reports/tables/survival_km_summary.csv`).

Variant	Events	Censoring	Median	S(24mo)	RMST 60mo
Broad	927	70.7%	undefined	0.880	50.5 mo
Balanced	**618**	**80.4%**	**undefined**	**0.914**	**53.5 mo**
Strict	309	90.2%	undefined	0.960	56.9 mo

14.2 Kaplan–Meier by CPV Division

Figure 13 stratifies by the four core digital CPV divisions under both the broad and strict variants. Ranking is *not* stable across variants: division 32 (telecommunications equipment) ranks fastest-recurring (lowest RMST, 54.7 mo) under broad but slowest under strict, and the Spearman correlation between the broad and strict RMST rankings over the four core divisions is $\rho = -0.40$ ($p = 0.60$, $n = 4$; `reports/tables/m6c_km_rank_correlation.csv`) — with only

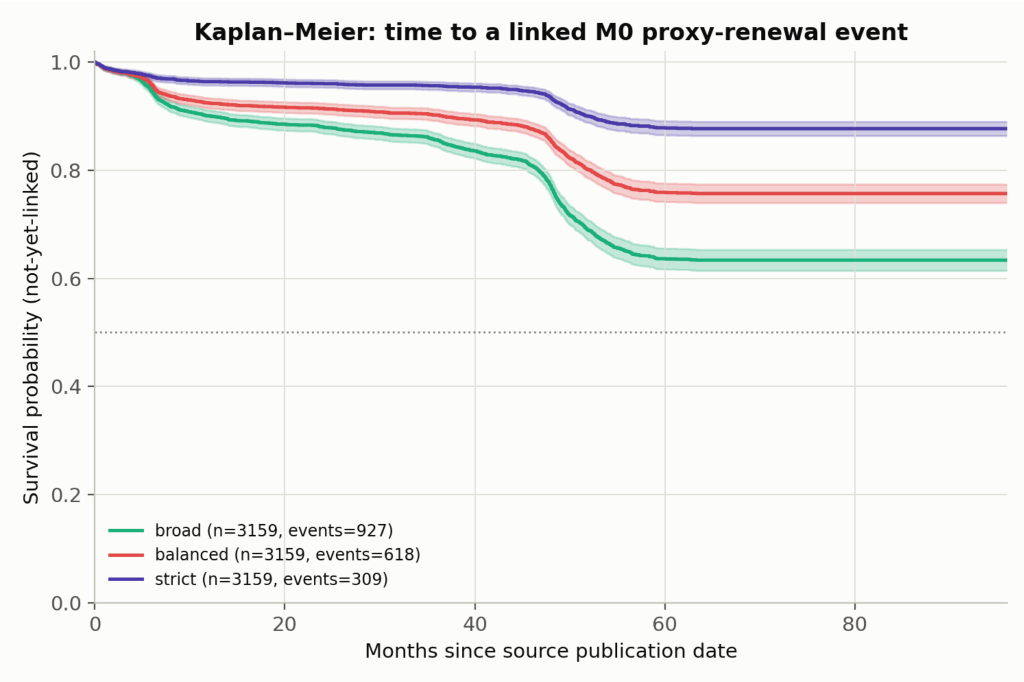

Figure 12: Kaplan–Meier survival curves (time to a linked M0 proxy-recurrence event) for the three named threshold variants, with pointwise 95% confidence bands. The dotted line marks the 50% level, which no curve reaches within the observed follow-up.

four divisions this test has essentially no power, but the observed *negative* point estimate is enough to rule out treating the ranking as stable. Division 72 (IT services) — the largest single division by volume (889 sources) — ranks consistently mid-to-low. A formal pairwise log-rank comparison across CPV divisions was attempted in this run but produced no usable output rows (a type-mismatch between the survival file's numeric-like `cpv_division` values and the comparison script's string literals, an implementation defect flagged here rather than silently worked around); the KM/RMST comparison below is therefore the descriptive substitute for a formal test in the current run.

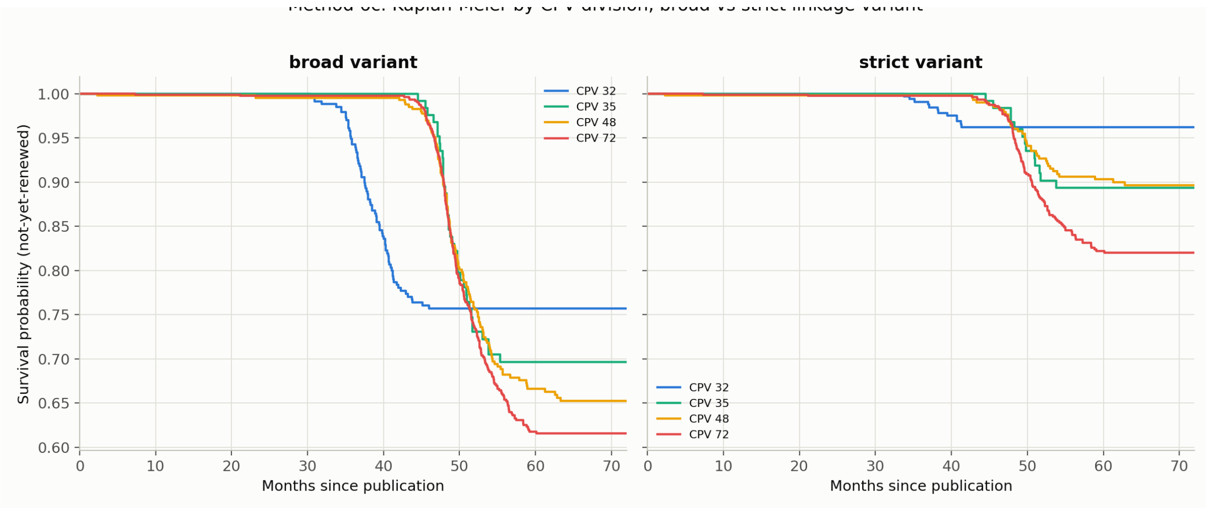

Figure 13: Kaplan–Meier by CPV division, broad (left) vs. strict (right) variant. Ranking instability across variants argues against confirmatory segment-level claims.

14.3 Cox Proportional Hazards Model

Table 22 reports the interpretable, non-CPV terms of the buyer-clustered Cox model fit on the balanced reference event definition ($n = 3{,}159$, 618 events); Figure 14 plots the same terms as hazard ratios. The full model also includes 31 CPV-division dummy terms (`reports/tables/survival_cox_models.csv`); several fire with extreme coefficients on sparse divisions (e.g. division 33 with only 10 sources, HR$= 21.3$, or division 43 with a single source, HR≈ 0), a documented symptom of separation in small strata rather than a credible substantive effect — this is itself a finding (Section 11, L7), not a result to be read at face value.

Table 22: Balanced clustered Cox model, non-CPV terms (`reports/tables/survival_cox_models.csv`).

Term	HR	95% CI	p
$\log(1 +$ declared duration, months)	0.923	[0.771, 1.105]	0.384
Duration was imputed (ref. observed)	**0.433**	[0.280, 0.670]	<**0.001**
Buyer key = RAW_SIRET (ref. NAME_FALLBACK)	1.224	[0.631, 2.375]	0.549

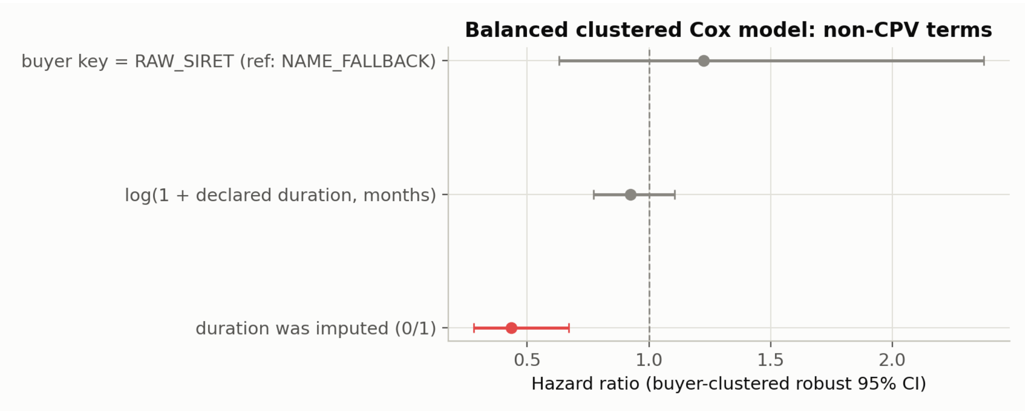

Figure 14: Forest plot of the balanced clustered Cox model's non-CPV hazard ratios (buyer-clustered robust 95% CI). Red marks the term significant at 5%.

The only term reaching conventional significance is `dur_was_imputed`: sources whose duration had to be imputed show a *lower* instantaneous linking hazard (HR$= 0.433$) than sources with an observed duration. Given that imputation defaults to a CPV-division median — which mechanically widens or shifts the $\pm$6-month blocking window relative to a source's true (unknown) contract length — this is at least as plausibly a *linkage-design artifact* as a substantive difference in recurrence behavior, and should not be reported as a business finding without the leakage caveats of Section 15.3. Declared duration itself is small and not significant in this design (HR$= 0.923$, $p = 0.38$); Section 15.3 shows this coefficient is unstable in sign and magnitude across alternative linkage designs.

14.4 Proportional-Hazards and Functional-Form Diagnostics

The Schoenfeld residual test on the balanced model flags four terms at $p < 0.05$: `log_duration` ($p = 6.8 \times 10^{-6}$), `dur_was_imputed` ($p = 7.7 \times 10^{-4}$), `cpv_division_72` ($p = 7.3 \times 10^{-5}$), and `cpv_division_MISSING` ($p = 0.042$) (`reports/tables/survival_ph_diagnostics_balanced.csv`). The duration-related flags are consistent with the functional-form check: adding a quadratic $(\log(1 + d))^2$ term to the clustered Cox model gives HR$= 1.010$ ($p = 0.365$) for the quadratic term itself while the linear term drops to HR= 0.992 ($p = 0.899$) — i.e. once duration and the duration-imputation flag are both in the model, most of the apparent duration signal is not well described by a simple linear log-hazard term, and should be read as an average effect over the observation window rather than a constant month-by-month multiplier.

14.5 Parametric AFT Models

Table 23: Parametric AFT model comparison, balanced reference (`reports/tables/survival_aft_models.csv`).

Model	AIC	Log-likelihood	Concordance
Weibull AFT	8,059.8	−3,990.9	0.7135
Log-normal AFT	**7,948.96**	**−3,935.5**	**0.7248**

The log-normal AFT model both fits better (lower AIC) and discriminates slightly better (concordance 0.725 vs. 0.714) than the Weibull AFT model, consistent with a non-monotone hazard shape rather than the Weibull's purely monotone one. No equivalent concordance figure was computed directly for the semi-parametric Cox model in this run (only for the two AFT models and, separately, the deferred temporal-validation attempt below); the AFT concordance is reported here as the best available in-sample discrimination summary and should not be read as an out-of-sample estimate.

14.6 Event Complexity and Temporal Validation

With 618 events, 3,159 sources, and 790 unique buyers, the compact Cox specification (duration + imputation + CPV + buyer-key-type dummies) has approximately 18.7 events per estimated parameter for the balanced variant — the strict variant, with only 309 events, has about 9.4, a level more vulnerable to overfitting and consistent with the separation symptoms noted in Section 14.3. A train-on-≤2021/test-on->2021 temporal validation split was attempted for every variant but explicitly deferred: `reports/tables/survival_temporal_validation.csv` records, for each variant, "deferred: sparse high-dimensional Cox design did not yield a stable temporal-validation fit in this audit run." This is reported transparently rather than silently omitted; a future validation design should collapse sparse CPV divisions, pre-specify a smaller covariate set, and avoid a split in which the test period is nearly entirely censored.

15 Sensitivity and Robustness

15.1 Alternative Event Definitions

Beyond broad/balanced/strict, three duration/window-design alternatives were built end-to-end (candidate generation through survival handoff), not as a post hoc filter of the existing 618 links:

- **No temporal score, reference pool**: keep the same 6-month duration-centered candidate pool, drop s_{time}, renormalize the remaining weights, rerank, rebuild links.
- **Forward 24-month, no duration**: generate candidates in a broad 24-month forward window from the source publication date, without centering on declared/imputed duration at all; score using text, CPV, and buyer reliability only.
- **Window 9/12/18 months**: rebuild the full candidate universe with a wider, still duration-centered, window.

Table 24: Overall survival summaries under all event definitions (`reports/tables/survival_km_summary.csv`).

Event definition	Events	Censoring	S(24mo)	RMST 60mo
Balanced reference	618	80.4%	0.914	53.5 mo
Broad	927	70.7%	0.880	50.5 mo
Strict	309	90.2%	0.960	56.9 mo
No temporal score, reference pool	618	80.4%	0.921	53.8 mo
Forward 24mo, no duration	1,105	65.0%	0.637	41.1 mo
Window 9mo	693	78.1%	0.905	52.8 mo
Window 12mo	754	76.1%	0.899	52.2 mo
Window 18mo	834	73.6%	0.890	51.4 mo

15.2 Temporal-Window Sensitivity

Rerunning candidate generation, scoring, ranking, acceptance, and survival construction end-to-end for $W \in \{6, 9, 12, 18\}$ months shows that a wider window recovers more sources into Ω but changes *which* sources are linked, not just how many:

Table 25: Temporal-window sensitivity (`reports/tables/m6d_temporal_window_sensitivity.csv`). Jaccard and status changes are relative to the 6-month reference.

Window	Sources w/ candidates	Pairs	Links	Coverage	Jaccard	Status changes
6 mo (reference)	1,236	6,137	618	39.1%	1.000	0
9 mo	1,386	8,692	693	43.9%	0.736	97
12 mo	1,508	10,793	754	47.7%	0.579	196
18 mo	1,668	14,097	834	52.8%	0.453	306

At 18 months, coverage improves by 13.7 points but nearly half the 6-month link set no longer overlaps (Jaccard 0.453), so the 6-month reference window is part of the scientific event definition, not a harmless implementation detail. Figure 15 visualizes the trade-off: blocking coverage and link counts rise monotonically with W while link-set overlap with the reference decays, and the downstream KM summary $\hat{S}(24)$ drifts from 0.914 ($W = 6$) to 0.890 ($W = 18$).

15.3 Duration Leakage and Circularity

Declared/imputed duration enters the pipeline three times: it constructs $\hat{e}_i$ (used for blocking), it drives s_{time} (used for scoring), and it remains available as a Cox/AFT covariate. This creates a genuine circularity risk that was tested directly, not assumed away.

The sign of the duration effect **reverses** between the duration-centered designs (HR < 1) and the duration-independent forward window (HR > 1), as Figure 16 makes visually explicit. This is strong evidence that declared duration should *not* be interpreted as an independent substantive predictor of recurrence under the current duration-centered linkage design — the apparent effect in Table 22 is at least partly a mechanical artifact of how candidates are searched for, not a free-standing finding about procurement behavior. This is the single highest-priority caveat in this report.

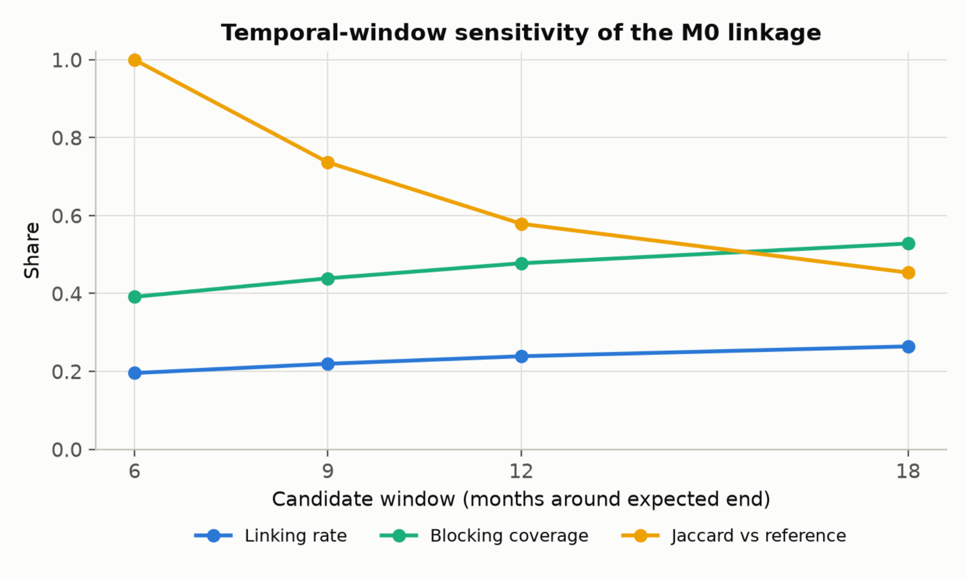

Figure 15: Temporal-window sensitivity of the M0 linkage (generated inline by notebook 14 from `m6d_temporal_window_sensitivity.csv`): blocking coverage and linked-event count increase with the window half-width W, while the Jaccard overlap with the 6-month reference link set falls to 0.453 at $W = 18$ — widening the window does not just find *more* links, it finds *different* ones.

Table 26: Duration-leakage linkage sensitivity (source: `reports/tables/duration_leakage_linkage_sensitivity.csv`).

Definition	Links	Event rate	Jaccard	Status changes	Median gap
Balanced reference	618	19.6%	1.000	0	37.7 mo
No temporal score, reference pool	618	19.6%	0.414	246	42.6 mo
Forward 24mo, no duration	1,105	35.0%	0.064	901	5.8 mo

Table 27: Duration-only clustered Cox audit across designs (source: `reports/tables/duration_leakage_cox_duration_effect.csv`); a single-covariate model isolating $\log(1 + d)$'s marginal association.

Design	HR (log dur.)	p (clustered)	Direction
Balanced reference	0.620	2.1×10^{-12}	Longer duration → lower hazard
No temporal score, reference pool	0.698	3.8×10^{-5}	Longer duration → lower hazard
Forward 24mo, no duration	**1.344**	9.0×10^{-8}	**Longer duration → *higher* hazard**

15.4 Quantitative Survival-Sensitivity Conclusion

The quantitative survival conclusion is therefore deliberately modest. Under the M0 balanced reference, 618 of 3,159 sources are linked to a constructed proxy-recurrence event, with 80.4% censoring, no reached median survival, $\hat{S}(24 \text{ months}) = 0.914$, and $\text{RMST}_{60} = 53.5$ months.

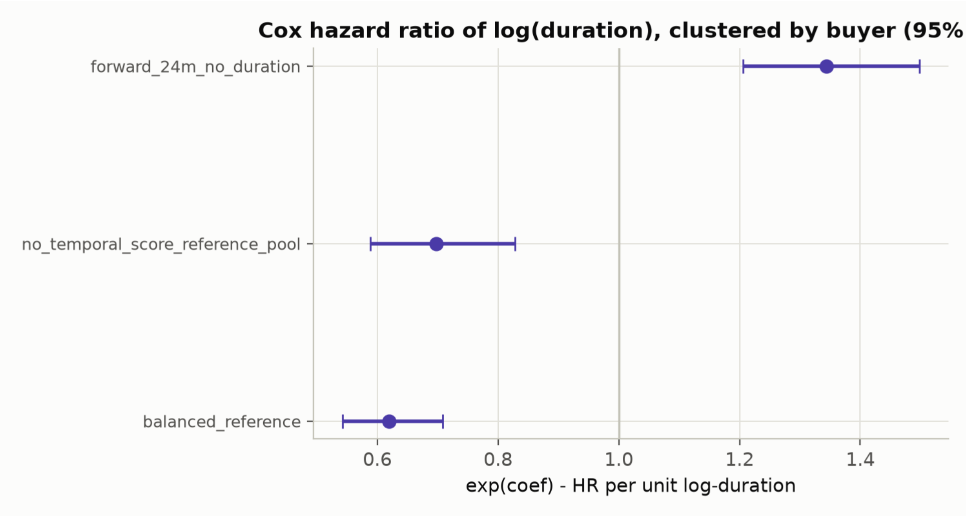

Figure 16: Buyer-clustered hazard ratio of $\log(1 + \text{duration})$ with 95% CI under the three linkage designs (generated inline by notebook 15 from `duration_leakage_cox_duration_effect.csv`). The confidence intervals of the duration-centered designs (HR < 1) and the duration-independent forward-window design (HR > 1) do not overlap — the effect's sign is an artifact of the design, not a stable property of the data.

Because the Kaplan–Meier curve does not cross 50%, RMST_{60} is the appropriate descriptive survival summary, not median survival.

The temporal-window sensitivity is moderate in the survival summaries but large in the underlying event identities. Moving from the 6-month reference window to an 18-month window raises the proxy-event count from 618 to 834 and the event rate from 19.6% to 26.4%, while RMST_{60} falls from 53.5 to 51.4 months. However, the link-set Jaccard overlap falls to 0.453 and 306 source event statuses change. The wider-window curve is therefore not simply a more complete version of the reference curve; it is based on a materially different constructed event.

The duration-design checks are more severe. Removing the temporal score while keeping the reference candidate pool preserves the event count at 618 but changes 246 source event statuses and lowers link-set overlap to 0.414. Replacing the duration-centered design with a forward 24-month no-duration window produces 1,105 events, lowers overlap to 0.064, changes 901 source event statuses, and reduces the median event time from 37.7 to 5.8 months. In the duration-only Cox audit, the clustered hazard ratio for $\log(1 + \text{duration})$ is 0.620 in the balanced reference and 0.698 in the no-temporal-score reference-pool design, but reverses to 1.344 under the forward 24-month no-duration design.

Finally, among parametric AFT models fit to the balanced reference, the log-normal model fits better than the Weibull model (AIC 7,949.0 vs. 8,059.8; concordance 0.725 vs. 0.714). This model comparison is still conditional on the constructed M0 event and should not be interpreted as validation of true renewal timing.

In short, the survival results are directionally informative but not definitive. They should be reported as robustness diagnostics for a constructed BOAMP proxy-recurrence event, not as stable evidence of verified renewal timing or as causal evidence about duration.

15.5 Prediction Calibration (12/24-Month Risk Groups)

As a bounded, non-operational analogue of a renewal-risk indicator, the fitted balanced Cox model's predicted risk at 12 and 24 months was grouped into quintiles and compared to the observed proxy-event rate within each quintile (`reports/tables/survival_predictions_12_24m.csv`).

Table 28: 12-month risk-quintile calibration, balanced variant.

Risk quintile (1=lowest)	n	Mean predicted risk	Observed event rate
1	632	0.029	0.002
2	632	0.041	0.008
3	631	0.057	0.003
4	632	0.082	0.082
5 (highest)	632	0.190	0.275

The highest-risk quintile does show a materially higher observed rate than the lower quintiles (Figure 17), so the model carries *some* ranking signal even in a design known to have duration-leakage concerns, but calibration is uneven (quintiles 1–3 under-predict relative to their already-small observed rates; quintile 5 under-predicts the true rate by roughly 8.5 points).

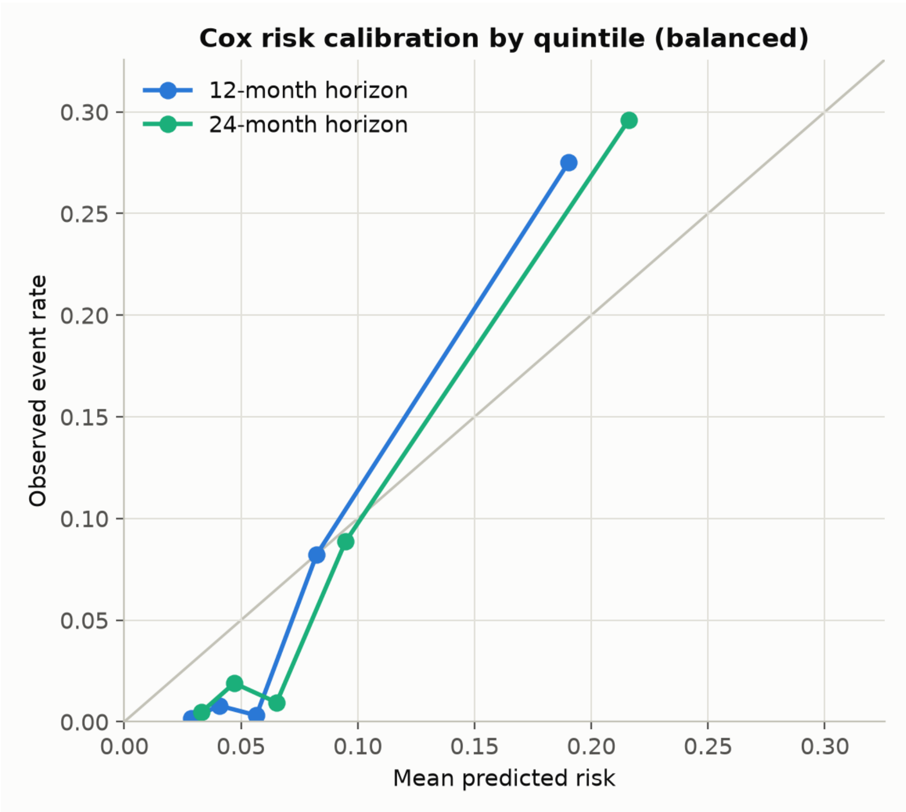

Figure 17: Cox risk calibration by predicted-risk quintile, balanced variant, 12- and 24-month horizons (generated inline by notebook 16 from `survival_predictions_12_24m.csv`). Bars compare mean predicted risk to the observed proxy-event rate within each quintile.

This is calibration against the constructed BOAMP proxy event, not against verified legal renewal, and given Section 15.3 it should not be operationalized as a renewal-risk score without first re-deriving it from a duration-independent linkage design and validating against manual labels.

16 Manual-Validation Readiness

No completed manual labels exist in this repository, and none were fabricated for this report. A stratified, unlabeled sample was drawn to `reports/tables/manual_validation_sample_unlabeled.csv`, covering score tiers, margin tiers, selected and non-selected pairs, both buyer-key types, imputed and observed durations, single- and multi-candidate sources, and duplicate-candidate cases. The accompanying guide (`reports/manual_validation_guide.md`) specifies four labels — *credible_renewal*, *likely_not_renewal*, *uncertain*, *insufficient_evidence* — and instructs reviewers not to infer a label from the M0 score. At least two independent reviewers and an adjudication step are recommended before treating labels as observed validation evidence.

17 Implementation Status of Every Method

The user-facing distinction that matters for a handoff is not "mentioned somewhere" vs. "not mentioned" but *how far each method actually got*. Table 29 classifies everything this project has touched, based on executable code and current outputs (the evidence standard of the credibility audit). Note that `audit_credibility_implementation_matrix.csv` records statuses *as found at the start of the audit*; the table below reflects the current state, after the audit itself implemented several of the items that CSV marks "NOT IMPLEMENTED."

18 What Can and Cannot Be Concluded

Supported by current evidence:

1. The balanced M0 survival handoff is reproducible and internally coherent (Section 4).
2. Blocking, not scoring, is the dominant restriction on event discovery: 60.9% of eligible sources never enter the candidate universe.
3. The composite score is internally discriminative: accepted links score higher and separate more clearly from runner-ups than the ambient candidate pool (Table 11).
4. Text degradation is the strongest single-field corruption stressor among the fixed-pool corruption tests.
5. The event definition is materially sensitive to the temporal-window design (Jaccard as low as 0.453 at an 18-month window).
6. The estimated duration effect on survival hazard is **not** robust to duration-independent linkage redesign — it reverses sign.
7. Kaplan–Meier median survival is undefined under every tested event definition; RMST at 60 months is the appropriate summary statistic.

Not supported by current evidence:

1. This pipeline does not establish verified legal renewal accuracy; `event`=1/0 are proxy-linking outcomes, not adjudicated facts.
2. The Fellegi–Sunter recall estimate is conditional on the blocked candidate universe Ω, not recall over all possible true renewals.
3. CPV-division segment rankings are not stable across event definitions (broad vs. strict) and are not confirmatory.

Table 29: Implementation status, current repository state.

Status	Items
Implemented and verified (code + current outputs + internal consistency checks pass)	Retrieval/parsing/preprocessing/candidate generation/linkage/survival handoff (34/34 final-audit checks; byte-identical notebook/script outputs); blocking-coverage, zero-candidate, duplicate-candidate, score-distribution, and censoring-balance diagnostics; threshold sensitivity; feature ablation; temporal-window sensitivity ($W \in \{6, 9, 12, 18\}$); duration-leakage audit (two alternative designs, end-to-end); KM/RMST for all eight event definitions; clustered Cox, PH and functional-form diagnostics; Weibull and log-normal AFT; 12/24-month calibration tables.
Implemented, still requiring external validation (runs correctly; meaning depends on unverified assumptions)	The M0 event definition itself (no ground truth — L1); the Fellegi–Sunter mixture precision/recall estimates (model-based, conditional on Ω); corruption-recovery rates (strict links are not ground truth; pool held fixed); every survival covariate effect (proxy event; duration circularity).
Partially implemented	Manual validation: the stratified unlabeled sample and annotation guide exist; no labels have been collected. CPV pairwise log-rank test: attempted but produced no usable rows in the current run (type mismatch between numeric-like `cpv_division` values and string literals — documented defect); descriptive KM/RMST comparison stands in for it.
Attempted and explicitly deferred	Temporal (train-on-≤2021 / test-on->2021) validation of the Cox model: attempted for all eight variants; recorded as "deferred: sparse high-dimensional Cox design did not yield a stable temporal-validation fit" in `survival_temporal_validation.csv`.
Evaluated and not adopted (deliberate, documented)	Sentence-transformer text embeddings (CPU-impractical at the original national scale; TF-IDF kept for continuity — revisiting is a listed next step); fuzzy buyer-name matching (`rapidfuzz` is installed but unused; exact normalized-name matching only); the DILA XML archive as retrieval source (Opendatasoft API preferred); the stricter CPV-division-only source definition (1,882 sources; kept as a sensitivity figure, not the headline population).
Superseded	The entire Run 1 national 2024–2026 corpus and its results (raw files archived under `data/raw/boamp_national_2024_2026_archive/`; no Run 1 output is part of any current result).
Proposed, not implemented	Weight-perturbation sensitivity grid; placebo/negative controls (runner-up, date-shifted, cross-buyer); full reblocking corruption study; one-to-one/capped many-to-one matching sensitivity; full-pipeline or buyer-level bootstrap; collapsed-CPV Cox redesign; SIRENE-based buyer reconciliation (would revisit the no-enrichment scope); "Grand Ouest" geographic extension.

4. Declared duration should not be reported as an independent predictor of recurrence under the current duration-centered design.

5. The 12/24-month risk-group calibration is calibration to the constructed proxy event, not to legal renewal probability, and should not be used operationally without further validation.

19 Reproducibility, Output Files, and Remaining Work

19.1 Reproduction Commands

The primary workflow is notebook-first (Section 2): retrieval stays in scripts, everything downstream runs in the notebooks with results inline. From the repository root:

```
# 1. Retrieval + parsing (external source -> flat interim table): scripts
```

```
python3 scripts/download_boamp.py
python3 scripts/parse_boamp.py

# 2. Pipeline: preprocessing, candidate pairs, linkage + survival handoff
jupyter nbconvert --to notebook --execute --inplace \
  notebooks/05_preprocess_boamp_m0.ipynb \
  notebooks/07_build_m0_candidate_pairs.ipynb \
  notebooks/08_run_m0_linkage.ipynb

# 3. Evaluation and analysis notebooks
jupyter nbconvert --to notebook --execute --inplace \
  notebooks/11_eval_m3_fellegi_sunter_mixture.ipynb \
  notebooks/12_eval_m4_corruption_recovery.ipynb \
  notebooks/14_eval_m6d_temporal_window_sensitivity.ipynb \
  notebooks/15_eval_duration_leakage.ipynb \
  notebooks/16_survival_robustness.ipynb
```

The equivalent headless batch backend (identical logic, byte-identical key outputs — Section 2.2):

```
python3 scripts/preprocess_boamp_m0.py
python3 scripts/build_m0_candidate_pairs.py
python3 scripts/run_m0_linkage.py
python3 scripts/eval_m3_fellegi_sunter_mixture.py
python3 scripts/eval_m4_corruption_recovery.py
python3 scripts/eval_m6d_temporal_window_sensitivity.py
python3 scripts/eval_duration_leakage.py
python3 scripts/run_survival_analysis.py
```

Script-only steps (no notebook counterpart): `audit_current_project.py` (viewed through notebook 13's `RUN_SCRIPT` toggle), `eval_m6a_threshold_sensitivity.py`, `eval_m6b_feature_ablation.py`, `eval_m6c_variant_km_robustness.py`, the figure builders (`make_figures.py`, `make_evaluation_figures.py`, `make_report_figures.py` — figures 01–18; the `nb14_*/nb15_*/nb16_*` figures are written by the notebooks themselves), and the report-assembly helpers (`_build_schema_summary.py`, `_build_source_trace.py`, `_build_final_audit.py`). Each retrieval script is idempotent with respect to `data/raw` (raw files are never overwritten) and runs are logged to `reports/run_logs/run_log.md`.

19.2 Key Output Files

19.3 Remaining Work

- **Manual validation** is the single highest-priority next step: label the stratified sample (Section 16), with two reviewers and adjudication, then compute observed precision/recall.
- **Redesign the duration-independent linkage variant** as a candidate primary specification, given the sign-reversal finding in Section 15.3, rather than treating it only as a sensitivity check.
- **Collapse sparse CPV-division dummies** (or apply stronger regularization) before re-attempting temporal (train/test) validation of the Cox model.
- **Fuzzy or SIRENE-linked buyer-name reconciliation** to reduce name-fallback fragmentation (L4) — while remaining within the project's stated no-external-enrichment scope, or explicitly revisiting that scope decision.

Table 30: Primary output files referenced throughout this report.

File	Purpose
`data/processed/boamp_m0_sources.csv`	M0 digital-scope source population (Section 6).
`data/processed/boamp_m0_candidate_pairs.csv`	All blocked/scored candidate pairs, Ω (Section 7).
`data/processed/boamp_m0_links_broad.csv`, `..._balanced.csv`, `..._strict.csv`	Accepted links per variant.
`data/processed/boamp_survival_m0_balanced.csv`	Official survival handoff (Section 8).
`reports/tables/audit_*.csv`	Credibility-audit tables (Sections 9, 12).
`reports/tables/m3_*.csv`, `m4_*.csv`, `m6*.csv`	Fellegi–Sunter, corruption-recovery, and sensitivity tables.
`reports/tables/survival_*.csv`	KM, Cox, AFT, PH-diagnostic, and calibration tables (Sections 14–15).
`reports/tables/manual_validation_sample_unlabeled.csv`	Stratified sample awaiting manual review (Section 16).

- **Full reblocking corruption study**, one-to-one/many-to-one adjudication of reused candidates, negative controls (runner-up, date-shifted, cross-buyer placebos), and full-pipeline or buyer-level bootstrap uncertainty all remain deferred, as documented in the implementation matrix (`reports/tables/audit_credibility_implementation_matrix.csv`).
- **Fix the CPV-division log-rank pairwise test** (Section 14) so that a formal significance test, not only descriptive KM/RMST comparison, is available for segment-level claims.
- If more recurrence signal is later needed, apply the guide's "Grand Ouest extension" fallback (Brittany/Normandy/Centre-Val de Loire departments) — current volume did not require it for this run.

Conclusion

The current M0 pipeline is a transparent, fully reproducible, and honestly audited first linkage design for BOAMP procurement recurrence in Pays de la Loire's digital/ICT sector. Its balanced survival handoff (3,159 sources, 618 proxy-recurrence events) is internally coherent and its composite score is measurably discriminative. At the same time, low blocking coverage (39.1%), non-trivial candidate reuse, real sensitivity to the temporal-window design, and — most importantly — a demonstrated duration-circularity effect on the estimated survival hazard mean that this report's survival findings should be read as **robustness diagnostics for a constructed BOAMP proxy event**, not as verified renewal probabilities or causal claims about procurement behavior. The defensible conclusion is that M0 provides a useful, fully auditable proxy-recurrence signal for exploratory survival analysis and prioritization, provided any downstream use explicitly states its conditional nature and treats manual validation as the next required step before operational deployment.

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